

Towards a Unified Framework for Determining Conformational Ensembles of Disordered Proteins

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构建用于确定无序蛋白质构象集合的统一框架

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Abstract

Disordered proteins play essential roles in myriad cellular processes, yet their structural characterization remains a major challenge due to their dynamic and heterogeneous nature. Here we present a community-driven initiative to address this problem by advocating a unified framework for determining conformational ensembles of disordered proteins. Our aim is to integrate state-of-the-art experimental techniques with advanced computational methods, including knowledge-based sampling, enhanced molecular dynamics, and machine learning models. The modular framework comprises three interconnected components: experimental data acquisition, computational ensemble generation, and validation. The systematic development of this framework will ensure the accurate and reproducible determination of conformational ensembles of disordered proteins. We highlight the open challenges necessary to achieve this goal, including force field accuracy, efficient sampling, and environmental dependence, advocating for collaborative benchmarking and standardized protocols.

Introduction

Intrinsically disordered proteins and proteins including intrinsically disordered regions (IDPs) are prevalent in eukaryotic proteomes, comprising an estimated one third of the human proteome¹. These proteins exist as conformational ensembles of rapidly interconverting conformations². IDPs are implicated in a broad range of signaling and regulatory functions, such as signal transduction, gene expression, genome organization, cell-cycle control and the formation of biomolecular condensates³. Consequently, IDPs are central targets for biological and medical research.

A defining feature of IDPs is the link between the features of their conformational ensembles and their biological functions⁴⁻⁶. Understanding this relationship requires a robust framework for generating and analyzing conformational ensembles by combining computational and experimental techniques. In this perspective, we advocate a unified framework for generating, analyzing, and validating conformational ensembles of IDPs. This framework, developed through extensive discussions initiated during a workshop held in Prague in May 2023 (URL: <https://ml4ngp.eu/workshop-prague>), is structured around three key modules: (1) experimental observations informative of conformational ensembles, (2) computational generation of conformational ensembles consistent with the experimental information, and (3) validation and comparison of the resulting conformational ensembles. By integrating insights from different methodologies in structural and computational biology, our goal is to progress toward establishing standardized processes that support integrative studies of IDPs.

This modular framework (**Fig. 1**) underscores the importance of interdisciplinary collaboration in addressing the complexities of IDPs. By standardizing methodologies and establishing consensus within the scientific community, this framework seeks to enhance our understanding of the functions of IDPs, advance methodological rigor, and accelerate therapeutic discovery.

摘要

无序蛋白质在众多细胞过程中发挥着至关重要的作用，然而由于其动态且具有异质性的特性，对其结构进行表征仍是一项重大挑战。本文介绍一项由学术界共同推动的倡议，旨在通过倡导一个用于确定无序蛋白质构象集合的统一框架来解决这一问题。我们的目标是将前沿实验技术与先进计算方法相结合，这些方法包括基于知识的采样、增强型分子动力学以及机器学习模型。该模块化框架由三个相互关联的组成部分构成：实验数据采集、计算构象集合生成以及验证。通过对该框架进行系统性开发，将确保对无序蛋白质的构象集合进行准确且可重复的测定。我们重点指出了实现这一目标所面临的若干开放性挑战——包括力场精度、高效采样以及环境依赖性问题，并倡导开展协同基准测试与制定标准化操作规程。

引言

固有无序蛋白以及含有固有无序区域（IDPs）的蛋白质在真核生物蛋白质组中十分普遍，约占人类蛋白质组的三分之一¹。这些蛋白质以构象集合的形式存在，其构象之间存在快速的相互转换²。IDPs 参与广泛的信号转导与调控功能，例如信号转导、基因表达、基因组组织、细胞周期调控以及生物分子凝聚体的形成³。因此，IDPs 是生物学与医学研究中的核心靶点。

IDPs 的一个显著特征在于其构象集合的特征与它们的生物学功能之间存在关联⁴⁻⁶。要理解这一关系，需要一个 robust 的框架，通过结合计算与实验技术来生成和分析构象集合。在本文中，我们倡导一个用于生成、分析和验证 IDPs 构象集合的统一框架。该框架是基于2023年5月在布拉格举行的研讨会（URL: <https://ml4ngp.eu/workshop-prague>）期间开展的广泛讨论而开发的，其结构围绕三个核心模块展开：(1)可为构象系综提供信息的实验观测数据；(2)与实验信息相符的构象系综的计算生成；(3)对所得构象系综进行验证与比较。通过整合结构生物学与计算生物学领域多种研究方法所获得的洞见，我们的目标是逐步建立标准化研究流程，以支持对内在无序蛋白（IDPs）开展整合性研究。

该模块化框架（图 1）突显了跨学科协作在应对内在无序蛋白（IDPs）复杂性问题中的重要性。通过标准化研究方法并在科学界达成共识，该框架旨在深化我们对 IDPs 功能的理解、提升方法学严谨性并加速治疗药物的发现进程。

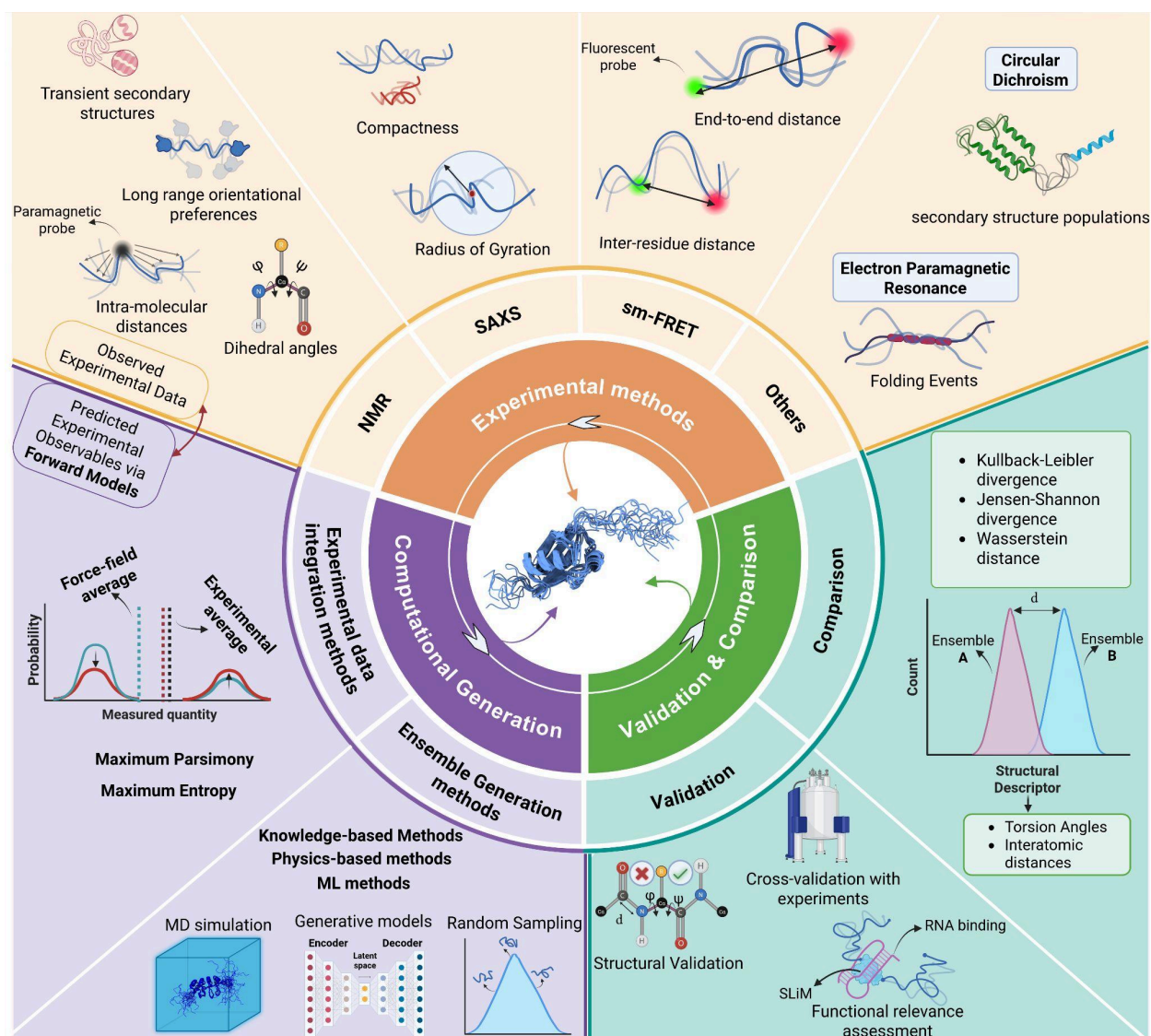


Fig. 1. Schematic representation of the modular framework for the determination of conformational ensembles of IDPs.

The illustration presents the modular framework for determining conformational ensembles of IDPs, comprising three interconnected modules: experimental methods, computational ensemble generation, and validation and comparison. The first module focuses on experimental techniques such as NMR spectroscopy, single-molecule FRET, SAXS, circular dichroism and electron paramagnetic resonance spectroscopy. These methods provide local and global structural information for characterizing the heterogeneous and dynamic nature of IDPs. The second module of the framework addresses the computational determination of conformational ensembles of IDPs. This module involves the generation of conformational ensembles and the integration of experimental data obtained in the first module. Ensemble generation methods can be categorized into physics-based (molecular dynamics simulations), knowledge-based, and machine-learning methods. In addition, maximum parsimony and maximum entropy are the main approaches for the integration of experimental data. The role of forward models is crucial in this step since they link the experimental observables to the generated models to ensure the goodness of fit to observed experimental data. Lastly, the validation and comparison tools complete the circular process of determining IDP conformational ensembles. Validation can be achieved through compliance of structures in the ensemble with basic physicochemical rules (structural validation), cross-validation using complementary experimental data, and functional relevance assessment. Comparison tools particularly involve statistical methods for analyzing distributions, such as Kullback-Leibler, Jensen-Shannon divergence scores and Wasserstein distances.

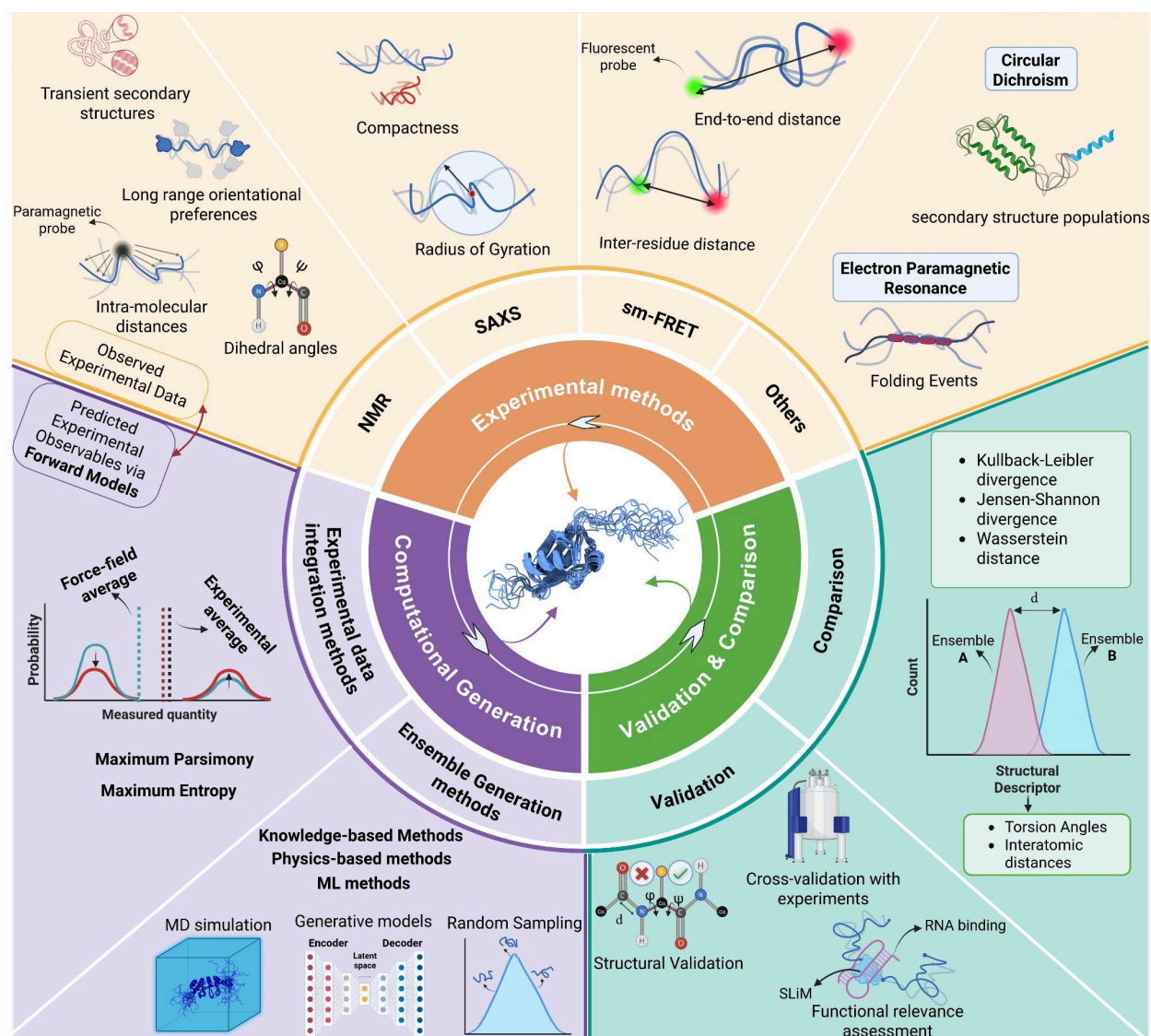


图 1. 用于构象测定的模块化框架示意图

无序蛋白集合。

该示意图展示了用于确定内在无序蛋白（IDPs）构象集合的模块化框架，该框架由三个相互关联的模块组成：实验方法、计算构象集合生成以及验证与比较。第一个模块侧重于实验技术，例如核磁共振波谱法、单分子FRET、小角X射线散射（SAXS）、圆二色谱法以及电子顺磁共振波谱法。这些方法可提供局部及全局的结构信息，用于表征内在无序蛋白（IDPs）的异质性与动态特性。该框架的第二个模块致力于通过计算手段确定IDPs的构象集合；该模块涉及构象集合的生成以及对第一模块中获取的实验数据进行整合。集成建模方法可分为基于物理原理的方法（如分子动力学模拟）、基于知识的方法以及机器学习方法。此外，最大简约法与最大熵法是整合实验数据的主要方法。前向模型在这一环节中发挥着至关重要的作用，因其将实验观测值与生成的模型相联系，从而确保模型对观测到的实验数据具有良好的拟合度。最后，验证与比较工具完善了确定IDP构象集合的闭环流程。验证可通过以下三种方式实现：一是检查系综中的结构是否符合基本物理化学规则（结构验证）；二是利用补充实验数据进行交叉验证；三是评估其功能相关性。比较分析工具尤其涉及用于分析分布的统计方法，例如Kullback-Leibler散度、Jensen-Shannon散度评分以及Wasserstein距离。

Box 1. Glossary of terms used within the scope of this paper

Term	Definition
Intrinsically disordered proteins and disordered protein regions (IDPs)	Full-length proteins or protein regions that lack a stable ordered three-dimensional structure under physiological conditions
Conformers	Structures able to interconvert without making or breaking covalent bonds
Conformational collection	A set of conformers
Statistical ensemble	A conformational collection endowed with statistical weights.
Conformational state	A thermodynamic state of a protein described by a statistical ensemble
Forward Model	A computational framework that predicts experimental observables based on protein structures

Experimental techniques for studying IDP structural ensembles

Nuclear magnetic resonance (NMR) spectroscopy

NMR spectroscopy is uniquely suited for studying IDPs in solution, offering atomic-level information. This technique can capture the dynamic and heterogeneous nature of IDPs by exploiting the sensitivity of nuclear magnetic moments to their local magnetic environment. This sensitivity enables precise, selective, and minimally invasive atomic-resolution assessment of structure, dynamics, and interactions of IDPs under near-physiological conditions⁷, typically following isotope labeling. As the free energy landscape is shallow for IDPs, NMR observables are time-averages over the different conformations adopted by the IDPs due to the relatively slow (e.g. milliseconds) NMR detection. As a result, NMR measurements typically accurately reflect the conformational state, in the sense that observables are representative of the statistical ensemble. However, the lack of the information on individual states can be viewed as a fundamental limitation. The ensemble-averaged NMR parameters do not enable the specific reconstruction of individual conformers of the ensemble. Instead, the NMR parameters can be compared to those derived by forward models from simulated structural ensembles and thus benchmark the latter. Also, there are cases when some states of IDPs can be distinguished (e.g., the presence of cis peptide bonds, which give rise to sets of additional signals^{8–10}), or when the NMR spectrum clearly carries a mark of their distribution (e.g., dynamics of molten globules manifested by line broadening¹¹). NMR can also unveil sparsely populated NMR-invisible structural states by exploiting the transfer of magnetization between these states

附录1：本文所用术语表

期限	定义
内在无序蛋白质与无序蛋白区域 (IDPs)	在生理条件下缺乏稳定有序三维结构的全长蛋白质或蛋白质片段
构象	能够在不形成或断裂共价键的情况下实现相互转化的结构
构象集合	一组构象异构体
统计系综	具有统计权重的构象集合。
构象状态	由统计系综描述的蛋白质热力学状态
前向模型	一种基于蛋白质结构预测实验可观测量的计算框架

研究IDP结构集合的实验技术

核磁共振（NMR）波谱学

核磁共振波谱法特别适用于研究溶液中的内在无序蛋白（IDPs），可提供原子级分辨率的信息。该技术通过利用核磁矩对局部磁环境的敏感性，能够捕捉IDPs所具有的动态性和异质性特征。这种敏感性使得在近生理条件下⁷对IDPs的结构、动力学及相互作用进行精确、选择性且微创的原子分辨率表征成为可能——此类表征通常在采用同位素标记后进行。由于IDPs的自由能景观较为平缓，加之NMR检测过程相对缓慢（例如仅需毫秒量级），因此NMR可观测量实际上是对IDPs所采取的不同构象进行的时间平均。由此，NMR测量结果通常能准确反映其构象状态——即这些可观测量代表了统计系综。然而，这种对单个构象状态信息的缺失可视为一个根本性局限。集合平均NMR参数无法用于对该集合中的单个构象异构体进行特异性重构。相反，可将这些NMR参数与通过前向模型从模拟结构集合中推导出的参数进行比较，从而对后者进行基准测试。此外，在某些情况下，可区分IDPs的不同状态（例如，存在顺式肽键，这会引发一组额外的信号⁸⁻¹⁰），或者当NMR光谱明显反映出这些状态的分布特征时（例如，通过谱线展宽体现的熔球态动力学¹¹）。核磁共振（NMR）还可通过利用这些NMR不可见结构态与NMR可见结构态之间磁化强度的转

and NMR-visible states, through techniques like saturation transfer difference (STD) NMR and relaxation dispersion (RD) NMR. By analyzing the transferred magnetization, these techniques provide structural and dynamic information about minor states populated during structural transitions upon binding¹². Moreover, NMR methods of selective detection of certain states have been designed (e.g. paramagnetic relaxation enhancement discussed below).

Key NMR observables in IDP studies

Several NMR parameters can be directly correlated to conformational properties of disordered proteins. **Chemical shifts** reflect local electronic environments and provide insights into secondary structure propensities (SSP)^{13–15}. Computational tools such as ShiftX¹⁶, SPARTA+¹⁷, PPM¹⁸, and CamShift¹⁹, include forward models to predict chemical shifts from conformational ensembles (Table 1).

J-couplings, particularly three-bond couplings, directly report on backbone torsion angles through the Karplus equation²⁰. These measurements are limited by the accuracy of the corresponding forward models and the availability of measurable J couplings for specific torsion angles. For example, ϕ angles can reliably be probed²¹. Furthermore, ψ angles often require complementary approaches such as cross-correlated relaxation^{22,23}.

Residual dipolar couplings (RDCs) provide highly sensitive insights into IDP conformational preferences by reporting on the average orientation of bond vectors across all coexisting structures in solution^{24,25}. RDCs reveal local secondary structure biases, having an opposite sign in alpha-helices and extended structures. However, their detailed structural interpretation requires calculating or fitting the alignment tensor, which is particularly challenging for IDPs. To avoid this calculation, forward models that use the laboratory reference frame have been proposed²⁶. Various alignment media exist, yet careful selection is crucial to avoid unwanted interactions that could alter measurements. Another use of RDCs is for the cross-validation of conformational ensembles derived from other NMR and SAXS data, ensuring consistency in structural interpretations²⁷.

Relaxation rates offer insights into IDP dynamics across timescales. Through IDP-specific model-free formalism, the analysis of several backbone ¹⁵N relaxation experiments in multiple magnetic fields probes the local and segmental motional amplitudes and their associated correlation times^{28,29}. These data are also valuable for benchmarking molecular dynamics (MD) simulations, in which ensemble reweighting may be required to match experimental relaxation rates³⁰.

Paramagnetic relaxation enhancement (PRE) is particularly powerful for disordered proteins, revealing transient long-range contacts and compact conformations within heterogeneous ensembles. For instance, PRE measurements have revealed transiently folded states in different IDPs^{31–33}, complementing global techniques like SAXS in characterizing conformational distributions^{34–36}. Paramagnetic relaxation interference (PRI) methods further expand this analysis, identifying correlated or anti-correlated motions in transiently compacted regions^{37,38}.

移——例如采用饱和转移差 (STD) NMR和弛豫色散 (RD) NMR等技术——来揭示稀疏存在的NMR不可见结构态。通过对转移磁化强度进行分析, 这些技术可提供关于在结合过程中发生结构转变时存在的次要结构态的结构与动力学信息¹²。此外, 已设计出用于选择性检测特定量子态的核磁共振 (NMR) 方法 (例如下文将讨论的顺磁弛豫增强技术)。

IDP研究中的关键核磁共振观测参数

若干核磁共振 (NMR) 参数可直接与无序蛋白质的构象特性相关联。**化学位移**反映了局部电子环境, 并有助于揭示二级结构倾向性 (SSP)^{13–15}。诸如 ShiftX¹⁶、SPARTA+¹⁷、PPM¹⁸以及 CamShift¹⁹ 等计算工具均包含用于根据构象系综预测化学位移的前向模型 (表1)。

J耦合 (尤其是三键耦合) 可通过 Karplus 方程²⁰直接反映主链扭转角。这些测量结果受相应前向模型的精度以及针对特定扭转角可测 J 耦合数据的可获得性所限。例如, ϕ 角可被可靠地探测²¹; 此外, ψ 角通常需要采用互补方法 (如互相关弛豫法) 进行测定^{22,23}。

残余偶极耦合 (RDCs) 通过报告溶液中所有共存结构中键矢量的平均取向, 为揭示IDP的构象偏好提供了高灵敏度的研究手段^{24,25}。RDCs可揭示局部二级结构偏好——在 α -螺旋和伸展结构中, 其耦合常数符号相反。然而, 要对RDCs进行详细的结构解析, 需计算或拟合对齐张量, 而这对于IDP而言尤为困难。为避免此类计算, 已有研究提出采用实验室参考系的前向模型²⁶。目前存在多种对准介质, 但需谨慎选择这些介质, 以避免可能影响测量结果的非预期相互作用。RDCs的另一项应用是对基于其他 NMR 和 SAXS 数据推导出的构象集合进行交叉验证, 从而确保结构解析结果的一致性²⁷。

弛豫率有助于揭示跨时间尺度下的非折叠蛋白 (IDP) 动力学行为。通过采用针对IDP设计的无模型形式体系, 对多种不同磁场条件下的主链¹⁵N弛豫实验进行分析, 可探究局部及区段性运动振幅及其对应的相关时间^{28,29}。这些数据对于分子动力学 (MD) 模拟的基准测试也具有重要价值; 在进行此类模拟时, 可能需要采用系综重加权方法以匹配实验测得的弛豫速率³⁰。

顺磁弛豫增强 (PRE) 对于无序蛋白质尤为有效, 能够揭示异质性体系中瞬时的长程相互作用及紧凑的构象。例如, PRE测量技术已揭示了不同无序蛋白 (IDPs) 中存在的瞬时折叠状态^{31–33}, 这为利用SAXS等全局性技术表征构象分布提供了补充^{34–36}。顺磁弛豫干涉 (PRI) 方法进一步拓展了这一分析方法, 可识别瞬时压实区域内的相关或反相关运动^{37,38}。

Since IDPs are highly sensitive to their environments, **in-cell NMR** is particularly indicated to study their context-dependent behavior under native-like conditions, such as in bacterial and eukaryotic cells. These studies, while technically challenging due to low sensitivity and high background noise, highlight the potential of NMR to bridge *in vitro* and *in vivo* IDP investigations^{39–42}.

Single-molecule Förster resonance energy transfer (smFRET)

smFRET is a powerful tool for investigating the conformational dynamics and structural heterogeneity of IDPs. FRET, a non-radiative energy transfer from an excited donor fluorophore to an acceptor fluorophore, is highly sensitive to donor-acceptor distances, following an inverse sixth-power relationship. This makes smFRET a spectroscopic ruler capable of measuring inter- and intramolecular distances within a 2–10 nm range⁴³. Unlike ensemble-averaging methods, smFRET provides a unique ability to resolve subpopulations, detect rare states, and capture dynamic transitions of individual molecules, making it particularly suited for IDP studies^{44,45}.

smFRET applications to IDPs

smFRET has been instrumental in studying IDPs under diverse conditions. It has been employed to elucidate the conformational ensembles of IDPs in dilute solution^{44,46,47} and crowded environments⁴⁸, including phase-separated biomolecular condensates^{49,50} and live cells⁵¹. Furthermore, this technique has shed light on dynamic IDP complexes, revealing the interplay of transient structural states during functional interactions⁴⁵. Single-molecule studies can be performed down to picomolar concentrations⁴³, avoiding the complications of aggregation and allowing functional/dysfunctional states of aggregation-prone IDPs to be probed^{52–54}.

smFRET experimental setup and protein labeling

smFRET experiments typically employ either Total Internal Reflection Fluorescence (TIRF) microscopy to observe immobilized molecules, or confocal spectroscopy to probe both freely diffusing and immobilized molecules. For IDPs, confocal-based smFRET offers several advantages, including the avoidance of perturbations from surface immobilization and the capability for multiparameter fluorescence detection, including wavelength, fluorescence lifetime, anisotropy, and correlation analyses⁵⁵.

Proteins are labeled site-specifically with donor-acceptor fluorophore pairs via cysteine residues or more advanced labeling strategies, such as non-natural amino acids or intein-based ligation methods^{56,57}. Heterogeneously labeled populations and different stoichiometries of intermolecular FRET can be resolved by alternating laser excitation (ALEX) or pulsed interleaved excitation (PIE)^{45,58}. The functional impact of labeling must be carefully assessed through control experiments, such as comparing the labeled and unlabeled IDP behavior in functional assays. Fluorophore mobility can be evaluated through time-resolved fluorescence anisotropy to ensure rotational freedom^{44,45,55}.

由于内在无序蛋白 (IDPs) 对所处环境高度敏感, **细胞内核磁共振 (NMR) 技术**尤其适用于研究这些蛋白在类天然条件 (例如细菌及真核细胞中) 下的环境依赖性行为。尽管由于灵敏度较低且背景噪声较高, 此类研究在技术上颇具挑战性, 但它们凸显了NMR技术在连接 *体外* 与 *体内* IDP研究方面的潜在价值³⁹⁻⁴²。

单分子Förster共振能量转移 (smFRET)

smFRET 是一种用于研究内在无序蛋白 (IDPs) 构象动力学及结构异质性的强大工具。FRET 是一种从激发态供体荧光团向受体荧光团进行的非辐射能量转移过程; 该过程对供体与受体之间的距离极为敏感, 其灵敏度遵循倒数六次方关系。因此, smFRET 可作为一种光谱学标尺, 用于测量 2–10 nm 范围内的分子间及分子内距离⁴³。与集合平均法不同, smFRET 具有独特的优势——能够解析亚群、检测罕见状态以及捕捉单个分子的动态转变, 因此特别适用于 IDP 研究^{44,45}。

smFRET技术在内在无序蛋白中的应用

smFRET 技术在研究不同条件下固有无序蛋白 (IDPs) 方面发挥了重要作用。该技术已被用于阐明 IDPs 在稀溶液^{44,46,47}以及拥挤环境⁴⁸ (包括相分离的生物分子凝聚体^{49,50}和活细胞⁵¹) 中的构象集合。此外, 该技术还揭示了动态 IDP 复合物的结构, 阐明了功能相互作用过程中瞬时结构状态之间的相互作用⁴⁵。单分子研究可应用于低至皮摩尔浓度的体系⁴³, 从而规避聚集现象带来的复杂性, 并能够对易聚集的内在无序蛋白 (IDPs) 的正常与异常功能状态进行探究⁵²⁻⁵⁴。

smFRET实验装置与蛋白质标记

smFRET 实验通常采用全内反射荧光 (TIRF) 显微镜技术来观察固定化分子, 或采用共聚焦光谱学技术来探测自由扩散及固定化分子。对于内在无序蛋白 (IDPs) 而言, 基于共聚焦技术的 smFRET 具有诸多优势, 例如可避免表面固定化带来的干扰, 以及支持多参数荧光检测 (包括波长、荧光寿命、各向异性及相关性分析) ⁵⁵。

蛋白质可通过半胱氨酸残基或更先进的标记策略 (例如非天然氨基酸或基于内含肽的连接方法^{56,57}) 进行位点特异性地用供体-受体荧光团对进行标记。通过交替激光激发 (ALEX) 或脉冲交错激发 (PIE) ^{45,58}, 可对异质性标记的分子群体以及不同化学计量比的分子间FRET现象进行分辨。标记所产生功能影响必须通过对照实验 (例如在功能测定中比较带标记与未标记的IDP行为) 进行仔细评估。荧光团的迁移率可通过时间分辨荧光各向异性法进行评估, 以确保其具有旋转自由度^{44,45,55}。

Quantitative analysis of smFRET data for IDPs

The FRET efficiency, E , for IDPs cannot be interpreted in terms of a fixed donor-acceptor distance. Instead, IDPs require analytical approaches that can account for the broad distributions, $P(r)$, of the inter-fluorophore distances r , and the dynamic time scales of protein and dye motions. Suitable models from polymer physics have commonly been employed for describing global IDP features and to approximate $P(r)$ ⁴⁶. Fluorescence lifetime measurements complement smFRET for accurate $P(r)$ determination. Two-dimensional plots of donor and acceptor fluorescence lifetime versus E can reveal whether a linear relationship exists (indicating fixed distances) or if deviations suggest broad and rapidly sampled distributions, offering a more stringent test of polymer models^{44,45}. The integration with advanced computational approaches, including molecular simulations and Bayesian reweighting, can provide more detailed descriptions of $P(r)$ and deviations from simple polymer models, for instance in the presence of folded parts^{47,59}. Nanosecond fluorescence correlation spectroscopy (nsFCS), combined with smFRET, enables the measurement of fast dynamics, including the reconfiguration times of disordered chains, which can be directly compared to simulations⁴⁶.

Important aspects for smFRET are to assess potential perturbations of the protein by fluorescent labeling, the analysis of complex distance distributions, and the role of the relevant timescales of the measurement for signal averaging^{46,60}. Advances in labeling technologies and integration with computational modeling, including an explicit representation of fluorophores, and complementary biophysical techniques make smFRET invaluable in integrative approaches of IDP ensemble determination^{61,62}.

Small-angle scattering (SAS) of X-rays and neutrons

Small-angle scattering (SAS) of X-rays (SAXS) and neutrons (SANS) have emerged as powerful techniques for the structural characterization of biomolecular systems in solution⁶³. Despite being a low resolution (1-2 nm) technique, SAS offers valuable structural information on the overall size and shape of biomolecules and biomolecular assemblies. In the context of IDPs, this information is key to constraint the ensemble description and it is complementary to residue-level information derived from other techniques^{64–66}. Major advances in beamline instrumentation, automatization, sample preparation and computational methods in the last decade have led to a tremendous increase in the application of SAS to structural biology⁶⁷. The fast readout capabilities of current detectors enable kinetic monitoring of reactions across a broad time scale, from μ s to hours, triggered by stopped-flow⁶⁸ or temperature jumps⁶⁹.

Although the conceptual framework of X-ray and neutron scattering as well as the strategies applied for their analysis are similar, these techniques differ in practical aspects, especially related to their sensitivity, which is much higher for SAXS^{70,71}. The advantage of SANS arises from the different scattering power of nuclei, particularly large between the proton and the deuteron. Exploiting this isotope effect through contrast variation experiments of partially deuterated samples has been a key concept to enrich the structural content of SANS experiments^{70,71}.

IDPs的smFRET数据定量分析

对于无序蛋白质 (IDPs)，其 FRET 效率 E 无法用固定的供体-受体距离来解释；相反，IDPs 需要采用能够考虑荧光团间距离 r 的广泛分布 $P(r)$ (22) 以及蛋白质与染料运动的动态时间尺度的分析方法。通常采用聚合物物理学中的合适模型来描述 IDPs 的整体特征并近似计算 $P(r)$ (22) ⁴⁶。荧光寿命测量可作为smFRET的补充手段，用于精确测定 r (22)。通过绘制供体与受体荧光寿命随 E 变化的二维图，可判断是否存在线性关系（表明分子间距离固定），或是否存在偏离现象（表明存在宽泛且快速采样的分布），从而对聚合物模型进行更严格的检验^{44,45}。与先进计算方法（包括分子模拟和贝叶斯重加权法）相结合，可对 ρ 及其与简单聚合物模型之间的偏差提供更详细的描述——例如在存在折叠结构的情况下^{47,59}。纳秒荧光相关光谱 (nsFCS) 与单分子FRET技术相结合，可实现对快速动力学过程的测量（包括无序链的重构时间），这些测量结果可直接与模拟结果进行对比⁴⁶。

smFRET 中的重要方面包括：评估荧光标记对蛋白质可能造成的扰动、分析复杂的距离分布，以及探讨测量中相关时间尺度在信号平均化中的作用^{46,60}。标记技术的进步及其与计算建模（包括对荧光团的显式表征）以及互补性生物物理技术的整合，使得 smFRET 在无序蛋白 (IDP) 系综测定的整合研究方法中具有不可替代的价值^{61,62}。

X射线与中子的小角散射 (SAS)

X射线小角散射 (SAXS) 与中子小角散射 (SANS) 已成为对溶液中的生物分子体系进行结构表征的有力技术⁶³。尽管该技术分辨率较低 (1–2 nm)，但 SAS 能为生物分子及生物分子组装体的整体尺寸与形状提供宝贵的结构信息。在内在无序蛋白 (IDPs) 研究中，这一信息对于约束整体结构描述至关重要，且与通过其他技术获得的残基水平信息^{64–66}相互补充。过去十年间，光束线装置、自动化技术、样品制备及计算方法方面的重大进展，推动了小角散射 (SAS) 在结构生物学领域应用范围的大幅扩展⁶⁷。当前探测器所具备的快速读出能力，使得能够对跨越从微秒到数小时这一宽时间尺度上的化学反应进行动力学监测，该监测过程由停流法⁶⁸或温度阶跃⁶⁹触发。

尽管X射线与中子散射的概念框架以及用于其分析的策略较为相似，但这两项技术在实际应用方面存在差异，尤其是灵敏度方面——SAXS的灵敏度要高得多^{70,71}。SANS的优势源于原子核不同的散射本领，尤其是质子与氘核之间的散射本领差异显著。通过对部分氘代样品进行对比度变化实验来利用这一同位素效应，已成为丰富SANS实验结构信息的关键概念^{70,71}。

Analysis of SAS data

The scattering profile of an IDP is the average of those arising from all coexisting conformations sampled in solution. As a consequence, SAS profiles measured for IDPs do not display features along the momentum transfer (q) range⁶⁴. Traditionally, Kratky plots ($I(q) \cdot q^2$ as a function of q), especially in their dimensionless form⁶⁵, have been used to qualitatively identify (partly) disordered states and distinguish them from globular particles. The Kratky representation has the capacity to enhance particular features of scattering profiles, allowing an easier identification of different degrees of compactness⁷².

The radius of gyration, R_g , which can be obtained from the smallest angles of a SAS curve using the Guinier approximation, is the most common descriptor to quantify the overall size of particles in solution. The experimental R_g is a single value representation of the size of the molecule, which for disordered states represents an ensemble average over all accessible conformations of the IDP and its hydration layer. The most common quantitative interpretation of R_g for IDPs is based on Flory's equation, which relates it to the length of the protein chain through a power law ($R_g \propto N^v$), where N is the number of residues and v is an exponential scaling factor. For chemically denatured proteins a v value of ≈ 0.6 is expected⁷³. However, according to theoretical models, IDPs are slightly more compact, with a v value of 0.522 ± 0.01 ⁷⁴. Recently, an approach has been developed to simultaneously derive the R_g and the v value from a single SAXS profile⁷⁵, which has been used to connect the compactness of IDPs with their capacity to phase separate^{36,76}.

The pairwise distance distribution, $P(r)$, derived from the Fourier transformation of the SAS profile, provides a 1D representation of the molecular structure, and it can also be used to qualitatively identify protein flexibility. The $P(r)$ also provides the maximum intramolecular distance, D_{max} , among all coexisting conformers. However, in the context of disordered proteins, D_{max} is not a robust parameter and its quantitative interpretation is not recommended^{77,78}.

For IDPs, ensemble approaches are required, relying on two key ingredients: (i) accurate models of disordered states, and (ii) forward models with the capacity to robustly predict scattering properties including contributions from the solvent to the measured SAS data⁷⁹.

Other biophysical techniques

Other complementary experimental methods have also gained attention in the studying of IDP structural properties (**Supplementary Table 1**). **Hydrogen deuterium exchange (HDX)** relies on measuring the rate at which amide protons in the protein backbone exchange with deuterium in a heavy water solution, a process highly sensitive to structural fluctuations and hydrogen bonding. For IDPs, HDX (HDX-NMR) by NMR spectroscopy or in particular by mass spectrometry (HDX-MS) provides valuable insights into disorder-to-order transitions, transient secondary structure formation, and conformational flexibility, making it a complementary technique to other structural methods like NMR, smFRET, or SAS. However, given the high exchange rates of IDPs, careful control of experimental conditions, such as pH, temperature, and labeling times, becomes critical to ensure meaningful interpretation of the data⁸⁰.

SAS 数据分析

IDP 的散射谱图是溶液中采样到的所有共存构象所对应散射谱图的平均值。因此，对 IDP 测得的 SAS 谱图在动量传递 (q) 范围⁶⁴内并未显示出明显特征。传统上，Kratky 图 ($I(q) \cdot q^2$ 作为 q 的函数)，尤其是其无量纲形式⁶⁵，常用于定性识别（部分）无序态并将其与球状粒子区分开来。Kratky 表示法能够突出散射谱图中的特定特征，从而更便捷地识别不同紧密度⁷²。

回转半径 R_g 可通过利用 Guinier 近似法从 SAS 曲线的最小角度求得，是用于量化溶液中颗粒整体尺寸的最常用表征量。实验测得的 R_g 是分子尺寸的单一数值表征；对于无序状态而言，该值代表了对 IDP 及其水合层所有可及构象所进行的系综平均。对于无序蛋白质 (IDPs)，对 R_g 的最常见定量解释基于 Flory 方程，该方程通过幂律关系将其与蛋白质链的长度联系起来 ($R_g \propto N^{\delta}$)，其中 N 表示残基数量，而 δ 为指数缩放因子。对于化学变性蛋白质，预期 δ 值约为 0.6⁷³。然而，根据理论模型，IDPs 的结构略为紧凑，其 δ 值为 0.522 ± 0.01 ⁷⁴。近期，研究人员开发出一种方法，可从单个 SAXS 谱图⁷⁵中同时推导出 R_g 与 δ 值，该方法已被用于将 IDPs 的致密性与其发生相分离的能力联系起来^{36,76}。

通过 SAS 谱图的傅里叶变换得到的成对距离分布 $P(r)$ ，为分子结构提供了二维一维表征，亦可用于定性判断蛋白质的柔性程度； $P(r)$ 还给出了所有共存构象异构体之间最大的分子内距离 D_{max} 。然而，在蛋白质结构无序的情况下， D_{max} 并非一个稳健的参数，因此不建议对其进行定量解读^{77,78}。对于无序蛋白质 (IDPs)，需采用系综方法，该方法依赖于两个关键要素：(i) 精确的无序状态模型，以及 (ii) 具备稳健预测散射特性能力的前向模型——该模型能够可靠地预测包括溶剂对实测 SAS 数据所作贡献在内的散射特性⁷⁹。

其他生物物理技术

其他互补性实验方法在研究无序蛋白质 (IDP) 的结构特性方面也受到了广泛关注（附表1）。**氢-氘交换 (HDX)** 技术依赖于测量蛋白质主链中酰胺质子与重水溶液中氘发生交换的速率；这一过程对结构波动及氢键作用具有高度敏感性。对于内在无序蛋白 (IDPs)，通过核磁共振波谱法（尤其是质谱法，即 HDX-MS）进行的氢氘交换核磁共振 (HDX-NMR) 技术可为理解蛋白质从无序态向有序态的转变、瞬时二级结构的形成以及构象灵活性提供重要见解，使其成为核磁共振、smFRET 或 SAS 等其他结构解析方法的互补性技术。然而，鉴于 IDPs 的高交换率，对实验条件（如 pH 值、温度及标记时间）进行严格控制至关重要，以确保对实验数据进行有意义的解读⁸⁰。

Mass spectrometry-based techniques beyond HDX-MS, such as hydroxyl radical footprinting (e.g. fast photochemical oxidation of proteins), diethylpyrocarbonate (DEPC) labeling, and carboxyl or lysine-specific covalent modifications, provide complementary insights into IDP dynamics by irreversibly labeling solvent-exposed side chains on rapid timescales⁸¹. These methods capture transient and heterogeneous conformations that HDX may miss, offering residue-level resolution of side-chain exposure. Applications include detecting structural transitions during amyloid formation⁸², mapping interaction interfaces⁸³, and identifying microenvironment-sensitive regions^{84,85}. When combined with modeling, these approaches enhance our ability to characterize the dynamic ensembles and functional states of IDPs.

Circular dichroism (CD) spectroscopy is a common technique for characterizing protein secondary structures by measuring the differential absorption of left- and right-circularly polarized light. Traditionally calibrated for globular proteins, CD spectroscopy has faced challenges in analyzing IDPs due to their weak and featureless spectra. Recent advancements have introduced new IDP reference datasets^{86,87}, enhancing the ability of this technique to detect conformational changes under various conditions. Despite these improvements, precise secondary structure quantification of IDPs still remains a challenge.

Electron paramagnetic resonance (EPR) spectroscopy, combined with site-directed spin labeling (SDSL), is used to detect unpaired electrons and measure distance distributions between specific residues. It has emerged as a powerful technique for studying IDPs⁸⁸. SDSL-EPR is particularly useful for monitoring folding events and structural behavior, as it is not limited by protein size or system complexity⁸⁹. The technique has been successfully applied to study disease-associated IDPs such as α -synuclein and tau, providing valuable insights into their conformational ensembles and interactions with the environment^{88,90}.

Integrative experimental modeling

Integrative experimental approaches rely on combining complementary experimental data to determine conformational ensembles. An analysis of the Protein Ensemble Database (PED)⁹¹ shows that the most common hybrid approaches combine SAXS with NMR and/or smFRET. Notably, Gomes et al.⁹² used experimental NMR and SAXS data to construct the conformational ensembles for Sic1 and phosphorylated Sic1, and subsequently employed smFRET data as an independent validation criterion. In another recent study, Borthakur et al.⁹³ demonstrated that reweighted ensembles obtained from MD simulations using different force fields converge to highly similar ensembles when incorporating extensive experimental data from NMR and SAXS. In addition, other combinations can be useful, such as SAXS, smFRET, fluorescence correlation spectroscopy and dynamic light scattering, as shown by Borgia et al.⁹⁴. Such integrative studies illustrate how the information from different techniques is complementary.

Bridging Experiments and Theory: Forward models

A critical element in translating experimental data into conformational ensembles is to define procedures, referred to as forward models, to calculate experimental observables from protein

除 HDX-MS 之外的基于质谱技术（例如羟基自由基足迹法（例如蛋白质的快速光化学氧化法）、二乙基焦碳酸酯（DEPC）标记法以及羧基或赖氨酸特异性共价修饰法）通过在快速时间尺度上对溶剂可及侧链进行不可逆标记，为理解无序蛋白质（IDP）的动力学提供了互补性的见解⁸¹。这些方法能够捕捉到氢氘交换（HDX）技术可能遗漏的瞬时且异质性的构象，从而实现残基水平的侧链暴露度解析。其应用包括检测淀粉样蛋白形成过程中的结构转变⁸²、绘制相互作用界面图谱⁸³以及识别对微环境敏感的区域^{84,85}。当这些方法与分子动力学建模相结合时，可显著提升我们表征内在无序蛋白（IDPs）动态集合体及功能状态的能力。

圆二色谱（CD）光谱法是一种通过测量左旋与右旋圆偏振光的差值吸收来表征蛋白质二级结构的常用技术。尽管该方法传统上针对球状蛋白质进行校准，但由于内在无序蛋白（IDPs）的光谱较弱且缺乏特征峰，其在分析此类蛋白质时仍面临挑战。近期的研究进展引入了新的 IDP 参考数据集^{86,87}，增强了该技术在各种条件下检测构象变化的能力。尽管取得了这些进展，但对 IDP 进行精确的二级结构定量分析仍是一项挑战。

电子顺磁共振（EPR）波谱法与定点自旋标记（SDSL）相结合，可用于检测未成对电子并测定特定残基间的距离分布。该技术已成为研究内在无序蛋白（IDPs）⁸⁸的重要工具。SDSL-EPR 尤其适用于监测蛋白质折叠过程及结构行为，因其不受蛋白质大小或体系复杂性的限制⁸⁹。该技术已成功应用于研究疾病相关无序蛋白（如 α -突触核蛋白和tau蛋白），为揭示其构象集合及与环境的相互作用提供了宝贵见解^{88,90}。

综合实验建模

整合性实验研究方法依赖于结合互补的实验数据来确定构象集合。对蛋白质构象集合数据库（PED）⁹¹的分析表明，最常见的混合研究方法是将SAXS与NMR和/或smFRET相结合。值得注意的是，Gomes等人⁹²利用实验获得的NMR和SAXS数据构建了Sic1及磷酸化Sic1的构象集合，随后将smFRET数据作为独立的验证标准。在另一项近期研究中，Borthakur等人⁹³证明：当结合来自NMR和SAXS的大量实验数据时，通过使用不同力场从分子动力学（MD）模拟获得的重新加权集成集合会收敛于高度相似的集合。此外，其他组合亦具应用价值，例如SAXS、smFRET、荧光相关光谱法及动态光散射——这一点由Borgia等人⁹⁴证实。这类综合性研究揭示了不同技术所获取的信息是如何相互补充的。

实验与理论的桥梁：前向模型

将实验数据转化为构象系综的关键环节在于定义一系列称为“前向模型”的计算方法，用于根据蛋白质结构计算实验可观测量^{95,96}。针对IDPs的前向模型所面临的一个主要问题在于：大多数实验

structures^{95,96}. A major problem of forward models for IDPs is that most experimental methods provide data describing the average properties of proteins in solution, often over different timescales, rather than features of individual protein molecules, thus requiring a corresponding conformational averaging in the calculations^{95,96}. Forward models estimating the experimental observables from generated models enable fitting experimental data obtained from various sources with a representative collection of conformations, as outlined in Table 1. Since the use of forward models influences the fitting procedure, selecting the appropriate model requires detailed understanding and considerations⁹⁷. For instance, parameters of forward models need to be adjusted, ideally through self-consistent refinement against experimental datasets of both forward models and conformational ensembles^{97–99}. Ultimately, experimental observations are influenced by both systematic and random errors, which must be accounted for to produce accurate and precise models⁹⁶.

To consider errors and the inherent variability of experimental observables, different Bayesian inference frameworks have been suggested^{96,97}. Nonetheless, the development of more accurate and computationally efficient forward models and ML tools to account for different sources of error remains an active area of research.

Table 1. Main experimental methods to determine structural ensembles of IDPs

Technique	Experimental measurement	Forward Models / Computational tools / Theoretical models	Information Content	Structural interpretation (e.g. residue level vs. global)	System perturbation	Environment (In-vivo / in-vitro / in-cell)	Reference database
NMR	Chemical shifts (traditional, 13C-NMR, ...)	ShiftX, SHIFTS, Sparta, PPM1, CamShift	Local secondary structure propensity; ensemble-averaged deviations from random coil values. Limited	Dihedral angles, secondary structure population, dynamics	Isotope labelling	In-cell / in-vitro	bmrB.io
		SSP, RCI, d2D, Shiftcrypt					
	PRE, PRI	dipole-dipole equation+correlation function, DEER-PREdict, MMM	Ensemble-averaged distances between spin labels and nuclei; provides insight into transient, weak, or low-population interactions and structural heterogeneity.	Distances/regions with correlated folding/dynamics	Chemical attachment of a paramagnetic label		
		covariance/correlation analysis					
	Relaxation rates	Model to data: molecular dynamics simulation+correlation function analysis+weighting	Time constants for intramolecular and global motions; order parameters reflecting flexibility; reveals timescales of molecular motions.	Correlation times+weights/order parameters	Isotope labelling		
		Data to model: Spectral density mapping, model-free analysis, IMPACT					
	DOSY	Data to model: Stokes-Einstein equation Model to data: Kirkwood-Riseman equation, HYDROPRO	Global hydrodynamic radius; informs on chain expansion or compaction	Translation diffusion coefficient			
	J couplings	Karplus equation	Backbone dihedral angles (ϕ); detects transient secondary structure.	Dihedral angles			
	Cross-correlated	Data to Model: Correlation functions and	Dihedral angles based on correlated relaxation	Dihedral angles			

方法所提供的数据描述的是蛋白质在溶液中的平均性质（这些数据通常涵盖不同的时间尺度），而非单个蛋白质分子的特征；因此，在计算过程中需要进行相应的构象平均化处理^{95,96}。前向模型通过基于生成模型来估算实验可观测量，使得利用具有代表性的构象集合来拟合来自不同来源的实验数据成为可能（如表 1 所示）。由于前向模型的使用会影响拟合过程，因此选择合适的模型需要深入的理解与审慎考量⁹⁷。例如，正向模型的参数需要进行调整，理想的做法是结合正向模型及构象系综的实验数据集进行自洽优化⁹⁷⁻⁹⁹。最终，实验观测结果受到系统误差与随机误差的共同影响；为构建准确且精密的模型，必须对这些误差进行充分考虑⁹⁶。

为考虑实验观测值中的误差及固有变异性，学界提出了多种贝叶斯推断框架^{96,97}。尽管如此，开发更为精确且计算效率更高的正向模型以及用于处理各类误差来源的机器学习工具，仍是当前一个活跃的研究领域。

表1. 确定无序蛋白（IDPs）结构集合的主要实验方法

技术	实验测量	前向模型 / 计算工具 / 理论模型	信息量	结构解析（例如：残基水平与.....） 全球)	系统扰动	环境（体内/体外） 内嵌式)	参考数据库
核磁共振（NMR）	化学位移 (传统, ¹³ C-NMR, ...)	ShiftX、SHIFTS、Sparta、PPM1、CamShift	局部二级结构倾向性；与无规卷曲值的集合平均偏差。 有限	二面角、二级结构分布、动力学	同位素标记	体内/体外	bmr.io
		SSP、RCI、d2D、Shiftcrypt					
	PRE, PRI	偶极-偶极方程+相关函数, DEER-PREdict, MMM	自旋标记与原子核之间的集合平均距离；有助于揭示瞬时、微弱或低占据数的相互作用以及结构异质性。	距离/区域 相关联 折叠/动力学	化学试剂 附着于 a 顺磁性标记物		
		协方差/相关性分析					
	放 松 率	模型至数据：分子动力学模拟 + 相关函数分析 + 权重赋值	分子内与全局运动的时间常数；反映分子柔性的序参数；揭示了分子运动的时间尺度。	相关性 乘数+权重/排序 参数	同位素标记		
		建模数据：谱密度映射、无模型分析、IMPACT					
	多西	数据用于建模：斯托克斯-爱因斯坦方程；数据用于建模：Kirkwood-Riseman 方程、HYDROPRO	全局流体动力学半径；用于反映链的扩张或压缩情况	翻译扩散 系数			
	J耦合	Karplus方程	主链二面角 (ϕ) ；用于检测瞬态二级结构。	二面角			
互相关运算 d	数据到模型：相关函数与	基于相关弛豫的二面角	二面角				

	cross-relaxation	Karplus-like equation	processes between two couplings.				
	RDCs	Modeling alignment and calculation of the coupling	Bond vector orientation; indicates transient structural elements and long-range order	Mutual orientation	External alignment (mechanically stressed gels, alignment media)		
SAXS	Scattering intensity as a function of the angle	CRY SOL/WAXSiS Pepsi-SAXS/FoXS	Provides global structural information	Global shapes and sizes, low resolution structures	None required	in-vitro	sasbdb.org
SANS	Scattering intensity as a function of the angle	CRYSON/pepsi-SANS	Complementary to SAXS	Global shapes and sizes, low resolution structures	Deuteration	in-vitro	sasbdb.org
(sm)FRET	Transfer efficiency (fluorescence bursts or trajectories)	Förster equation + polymer models, molecular simulations + dye model (e.g. FRET Positioning and Screening, Seidel), rotamer-library approaches (e.g. FRETpredict)	Number of populations, FRET efficiencies, fluorescence lifetimes, fluorescence anisotropies	Intra- or intermolecular distance/distance distribution, conformational changes	Dye labeling	In-vitro/in-cell	N/A
	Fluorescence lifetime decay			Intra- or intermolecular distance distribution			
	Time correlation (nanosecond FCS)			Distance relaxation time			

Computational generation of conformational ensembles

Generating conformational ensembles through computational techniques is a critical step in integrative modeling of IDPs, providing atomic-level insight that complements experimental measurements. Recent progress has been driven by tighter integration between experimental and modeling efforts, where experimental data refine models and models aid in the interpretation of complex experiments⁹⁵. Three main strategies are commonly used to generate conformational ensembles: **knowledge-based methods**, which draw on empirical patterns or structural fragments; **physics-based simulations**, which are based on biophysical principles to sample the conformational landscape; and **machine learning-driven approaches**, which infer structural features from data without explicit physical modeling. Each strategy offers distinct advantages, and their selection depends on the level of detail required, available data, and modeling goals.

Knowledge-based methods

Knowledge-based methods (**Fig. 2a**) generate conformational ensembles using heuristics or statistical patterns derived from known protein structures. These approaches often rely on fragment libraries, rotamer statistics, or backbone angle distributions informed by structural

	交叉弛豫 n	Karplus类方程	两个耦合器之间的过程。				
	RDCs	耦合关系的建模与计算	键向量取向；用于表征瞬时结构单元与长程有序性	相互定向	外部对中 (机械式) 受力凝胶, 对准介质)		
小角X射线散射 (SAXS)	散射强度随角度变化的函数关系	CRY SOL/WAXSiS、Pepsi-SAXS/FoXS	提供全局结构信息	全局形状与尺寸、低分辨率结构	无需任何条件	体外	sasbdb.org
SANS	散射强度作为功能角度	CRYSON/百事可乐-SANS	作为 SAXS 的补充手段	全局形状与尺寸、低分辨率结构	氘代	体外	sasbdb.org
(sm)FRET	转移效率 (荧光脉冲或轨迹)	Förster方程 + 聚合物模型、分子模拟 + 染料模型 (例如 FRET定位与筛选法、Seidel 法)、旋转异构体库方法 (例如 FRETpredict)	群体数量、FRET效率、荧光寿命、荧光各向异性	腹腔内或分子间距离/距离分布、构象变化	染料标记	体外/细胞内	不适用
	荧光寿命衰减			腹腔内或分子间距离分布			
	时间相关性 (纳秒级FCS)			距离弛豫时间			

构象集合的计算生成

通过计算技术生成构象集合是蛋白质内源性无序蛋白 (IDPs) 整合建模中的关键步骤, 它可提供原子级层面的见解, 从而对实验测量结果形成补充。近年来, 这一领域的进展得益于实验研究与建模工作之间更紧密的整合——实验数据用于优化模型, 而模型则有助于对复杂实验结果进行阐释⁹⁵。生成构象集合通常采用三种主要策略: **基于知识的方法**——利用经验性模式或结构片段; **基于物理的模拟**——基于生物物理原理对构象景观进行采样; 以及**基于机器学习的方法**——无需显式物理建模即可从数据中推断结构特征。每种策略均具有独特的优势, 其选择取决于所需的细节程度、可用数据以及建模目标。

基于知识的方法

基于知识的方法 (图 2a) 利用从已知蛋白质结构中提取的启发式方法或统计模式来生成构象系综。这些方法通常依赖于片段库、旋转异构体统计量或基于结构数据库提供的主链角度分布信

databases. Examples of such methods include Flexible-Meccano¹⁰⁰, TraDES¹⁰¹, and IDPConformerGenerator¹⁰², which employ fragments from non-redundant structural databases and statistical coil models to generate ensembles. Because they do not require energy calculations, they are computationally efficient and suitable for generating large conformational pools. These conformational collections can then be refined or reweighted based on experimental constraints to approximate statistical ensembles. Importantly, knowledge-based approaches are particularly useful when only limited experimental data are available, or when rapid conformational collection generation is needed for large-scale applications. A summary of representative methods and typical use cases is provided in **Supplementary Table 2**.

Physics-based methods

Physics-based approaches (**Fig. 2b**) use molecular simulations to generate IDP ensembles. Since force fields originally developed for folded proteins often underestimate the flexibility and expanded conformations typical of disordered chains, IDP-specific force fields such as a99SB-disp¹⁰³ and Amber ff99SBws/ff03ws¹⁰⁴ have been developed to improve the balance between intramolecular and solvent interactions to better reflect IDP conformational preferences. Another challenge is sampling efficiency, as IDPs explore vast and shallow free energy landscapes. Techniques like replica exchange molecular dynamics (REMD)¹⁰⁵, which run simulations in parallel at different temperatures to enhance conformational exploration, are commonly used to improve sampling. More recently, hybrid strategies that combine simulation with machine learning (ML)—for example, using ML-derived potentials to guide sampling—have emerged as promising tools^{106,107}. Alternative frameworks such as hydrophobicity scale (HPS) models¹⁰⁸ and CALVADOS^{109,110}, coarse-grained models parameterized against experimental data on IDPs and multidomain proteins, reduce computational cost while preserving some of the essential sequence-dependent behaviors. Beyond ensemble generation, physics-based methods increasingly contribute to design-oriented frameworks—from modeling the thermodynamic basis of multiphase condensates¹¹¹, to uncovering sequence–ensemble–function relationships in live cells¹¹², and enabling inverse design of IDRs with target ensemble properties^{113–115}. Representative methods and applications are summarized in **Supplementary Table 2**.

Machine learning (ML) methods

ML approaches (**Fig. 2c**) have recently emerged as an alternative strategy for generating conformational ensembles of IDPs. These methods aim to learn sequence–ensemble relationships directly from structural data, providing a way to overcome some of the limitations of traditional simulation-based techniques. For example, idpGAN¹¹⁶ is a generative adversarial network (GAN) trained on ensembles generated by CG molecular dynamics simulations, capable of producing novel ensembles with realistic R_g . While still in early stages, ML-based models offer a promising route for efficient ensemble generation, especially when integrated with simulation-derived^{117,118} and/or experimental data¹¹⁹. In addition to generating ensembles, ML and related statistical parameter-learning methods can be used to tune force field

息。此类方法的示例包括 Flexible-Meccano¹⁰⁰、TraDES¹⁰¹以及 IDPConformerGenerator¹⁰²；这些方法利用来自非冗余结构数据库的片段及统计性卷曲模型来生成构象集合。由于这些方法无需进行能量计算，因此计算效率高，适用于生成大规模构象库。这些构象集合随后可根据实验约束条件进行精细化处理或重新加权，以近似得到统计系综。值得注意的是，当仅可获得有限的实验数据，或在需要为大规模应用快速生成构象集合时，基于知识的方法尤为适用。代表性方法及典型应用场景的汇总详见**补充表2**。

基于物理原理的方法

基于物理的计算方法（**图 2b**）利用分子模拟技术生成无序蛋白质（IDP）系综。由于最初为折叠蛋白质开发的力场往往低估了无序链所特有的柔性和扩展构象，因此开发了针对无序蛋白（IDP）特异性的力场（如 a99SB-disp¹⁰³和 Amber ff99SBws/ff03ws¹⁰⁴），旨在优化分子内相互作用与溶剂相互作用之间的平衡，从而更准确地反映 IDP 的构象偏好。另一个挑战在于采样效率，因为内在无序蛋白（IDPs）所处的自由能景观既广阔又浅层。诸如副本交换分子动力学（REMD）¹⁰⁵——该方法通过在不同温度下并行运行模拟来增强构象探索——等技术常被用于提升采样效率。近年来，将模拟方法与机器学习（ML）相结合的混合策略——例如利用基于机器学习推导出的势能来指导采样——已成为颇具前景的工具^{106,107}。替代性计算框架——例如疏水性标度（HPS）模型¹⁰⁸与CALVADOS模型^{109,110}（这些粗粒化模型是基于针对无序蛋白（IDPs）及多结构域蛋白质的实验数据进行参数化的）——在降低计算成本的同时，保留了部分与序列相关的本质行为。除集合生成方法外，基于物理原理的方法在面向设计的框架中发挥着日益重要的作用——从构建多相凝结物的热力学基础¹¹¹，到揭示活细胞中的序列-集合-功能关系¹¹²，再到实现具有目标集合特性的IDRs的逆向设计¹¹³⁻¹¹⁵。代表性方法与应用汇总于**补充表2**。

机器学习（ML）方法

机器学习方法（**图 2c**）近期已成为生成内在无序蛋白（IDPs）构象系综的一种替代性策略。这些方法旨在直接从结构数据中学习序列与系综之间的关系，为克服传统基于分子动力学模拟技术所存在的某些局限性提供了一种途径。例如，idpGAN¹¹⁶是一种基于CG分子动力学模拟生成的系综训练的生成式对抗网络（GAN），能够生成具有真实R_g值的新型系综。尽管仍处于早期阶段，基于机器学习的模型为高效生成系综提供了极具前景的途径，尤其是在与模拟衍生的^{117,118}及/或实验数据¹¹⁹相结合时。除了生成分子集合外，机器学习及相关统计参数学习方法还可用于根据实验数据

parameters against experimental data^{115,120}, predicting IDP properties directly from sequence^{4,6}, designing novel IDP variants with desired properties^{113,121} and predicting phase separation propensities of IDPs¹²². Representative ML-based methods for computational ensemble generation are summarized in **Supplementary Table 2**.

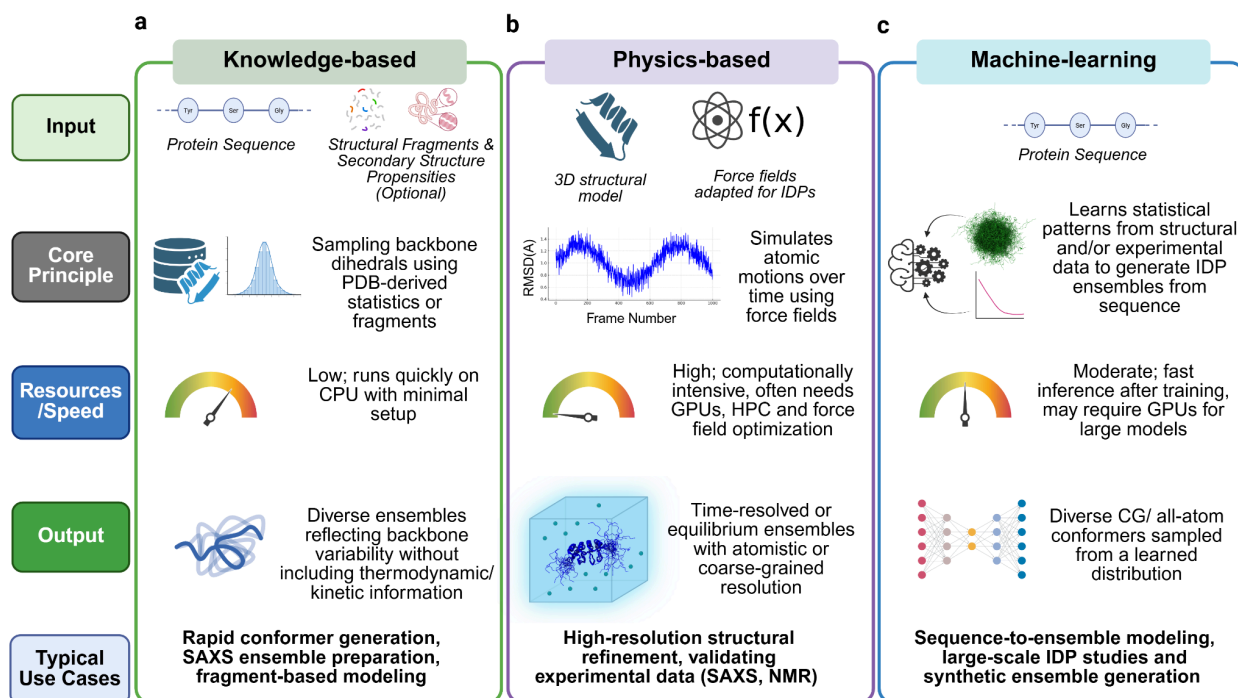


Fig. 2. Schematic comparison of computational strategies for the generation of conformational ensembles of IDPs. The diagram contrasts three broad methodological classes based on their inputs, underlying principles, computational demands, and typical applications. These approaches span from a) knowledge-based methods and b) physics-based simulations to modern c) machine learning techniques, each offering unique strengths for capturing IDP structural heterogeneity.

Integration of computational models with experimental data

Experimental techniques such as NMR, SAXS, and smFRET provide important insights into the behavior of IDPs, but these measurements typically report on averages over a wide range of rapidly fluctuating conformations. By contrast, computational methods—such as molecular simulations—can generate detailed conformational ensembles from a protein sequence. However, discrepancies often arise between simulations and experiments, reflecting inaccuracies in force fields, limited sampling, or the nature of the experimental observable itself

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To address this aspect, an integrative modeling strategy is needed—one in which experimental data and computational ensembles inform and refine each other. The basic workflow (**Fig. 3**) involves generating an ensemble using a physics-based, knowledge-based or machine learning model, computing experimental observables from that ensemble using forward models, and comparing them with actual experimental measurements. When agreement is poor, the ensemble can be refined or reweighted so that its average properties better match the data.

对力场参数进行调优^{115,120}、直接根据序列预测无序蛋白质（IDP）的性质^{4,6}、设计具有特定性质的新型IDP变体^{113,121}以及预测IDP的相分离倾向性¹²²。基于机器学习的代表性计算集成生成方法汇总于补充表2中。

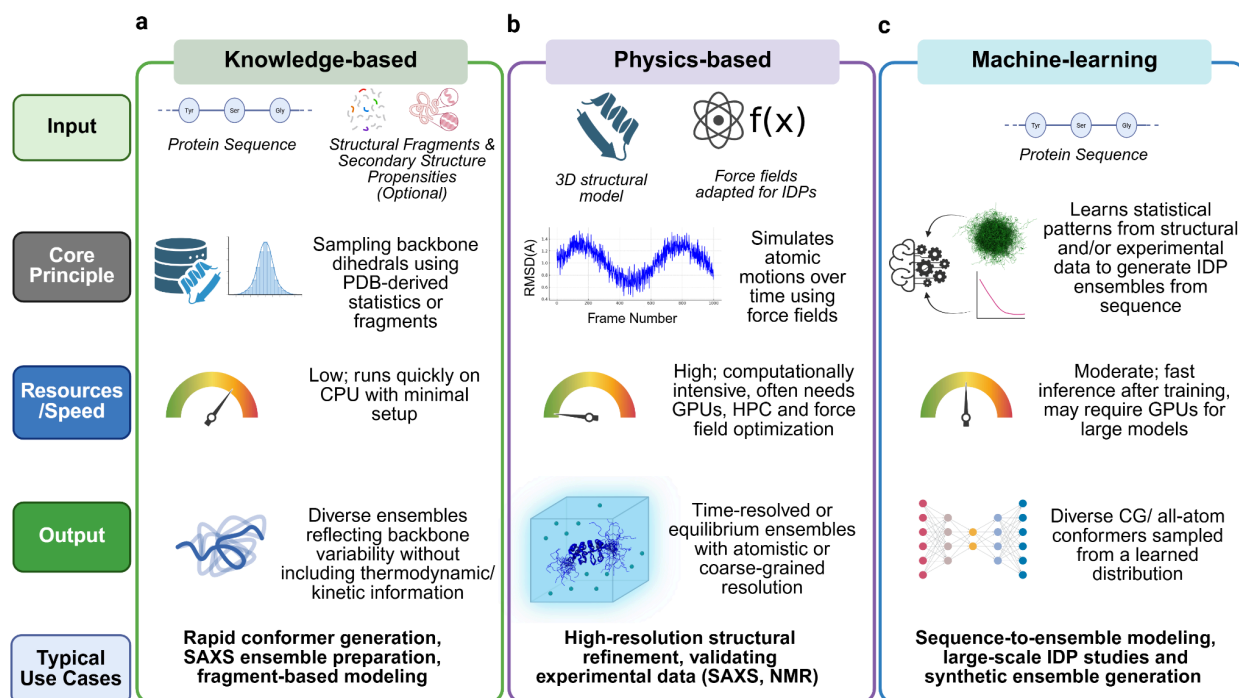


图 2. 用于生成 IDP 构象集合的计算策略示意图对比。该图根据输入数据、基本原理、计算需求及典型应用将三种广泛的方法学类别进行对比。这些方法涵盖从 a) 基于知识的方法到 b) 基于物理的模拟，直至现代 c) 机器学习技术；每种方法在捕捉 IDP 结构异质性方面均具有独特的优势。

计算模型与实验数据的整合

核磁共振（NMR）、小角X射线散射（SAXS）及单分子FRET（smFRET）等实验技术为理解无序蛋白（IDPs）的行为提供了重要见解；然而，这些测量结果通常反映的是对广泛快速波动构象所取的平均值。相比之下，计算方法（例如分子动力学模拟）能够根据蛋白质序列生成详细的构象系综。然而，模拟结果与实验结果之间常存在差异，这反映了力场的不精确性、采样范围有限或实验可观测量本身的特性¹²³。

为解决这一问题，需要采用一种整合建模策略——即让实验数据与计算集合相互启发、相互优化。该基本工作流程（图 3）包括：利用基于物理原理、基于知识或基于机器学习的模型生成计算集合；利用前向模型根据该计算集合计算实验可观测量；并将这些计算结果与实际实验测量值进行比较。当一致性较差时，可对集成模型进行精细化调整或重新加权，使其实现的平均特性更符合数据特征。

Two main classes of approaches are available for this purpose. The first is based on maximum entropy (ME)¹²⁴, a foundational concept in statistical mechanics for generating the most unbiased ensemble consistent with known constraints. In its simplest form, and in the absence of experimental data, structural ensembles follow the Boltzmann distribution: $P_i = \frac{1}{Z} e^{-\beta E_i}$ where E_i is the energy of conformation i , β is the inverse thermal energy, and Z is the partition function. However, when experimental observables are introduced, such as NMR chemical shifts or SAXS profiles, this framework is generalized to a Gibbs distribution: $P_i = \frac{1}{Z} e^{-\sum_k \beta_k E_{ik}}$. Here, E_{ik} represents an effective energy term for the deviation of conformation i from experimental observable k , and β_k (a Lagrange multiplier) adjusts the weight of that constraint. This formulation ensures the structural ensemble remains as close as possible to the original distribution while still fulfilling the experimental data. In practice, this statistical foundation, together with the principles from Bayesian statistics, underlies a number of ME-based methods such as Bayesian/Maximum Entropy (BME)¹²⁵, BioEn¹²⁶, and metainference⁹⁶. These approaches implement the principle either by reweighting existing conformers post hoc, or by applying biasing forces during simulations to guide sampling. Such techniques are especially effective for IDPs, whose native states are characterized by extensive structural heterogeneity that must be preserved during model refinement.

The second category, known as maximum parsimony (MP) methods, seeks to identify the smallest subset of conformers from a large pool that can adequately explain the experimental data, minimizing structural complexity. In contrast to ME, MP methods aim to simplify the conformational collection by selecting or reweighting structures to maximize agreement with experimental observations while minimizing the size of the conformational collection. These approaches, such as EOM¹²⁷ and ASTEROIDS¹²⁸, are particularly useful when only a few conformational states are dominant and are thus most suitable for low-entropy systems with limited conformational heterogeneity, where a description in terms of statistical ensemble is not required.

Altogether, these integrative strategies allow researchers to model structural ensembles that are both physically realistic and experimentally validated, bridging the gap between simulation and experiments in the study of disordered proteins. Representative techniques for integration of experimental data are provided in **Supplementary Table 3**.

为此可采用两大类方法。第一类方法基于最大熵（ME）¹²⁴——这一统计力学中的基础概念，用于生成符合已知约束条件且具有最大无偏性的系综。在最简单的情况下，且在缺乏实验数据时，结构系综遵循玻尔兹曼分布。： $P_i = \frac{1}{Z} e^{-\beta E_i}$ 其中， E_i 表示构象 i 的能量， β 为逆热能量， Z 为配分函数。

该函数。然而，当引入实验可观测量（例如 NMR 化学位移或 SAXS 谱图）时，该框架可推广为吉布斯分布。： $P_i = \frac{1}{Z} e^{-\sum_k \beta_k E_{ik}}$ 此处， E_{ik} 表示用于描述构象 i 偏离基准值的有效能量项。

实验可观测量 θ ，而 β （一个拉格朗日乘子）用于调节该约束条件的权重。 θ

该公式设计确保了结构集合尽可能接近原始分布，同时仍满足实验数据的要求。在实际应用中，这一统计学基础结合贝叶斯统计学原理，构成了多种基于最大熵方法的算法的基础，例如贝叶斯/最大熵（BME）¹²⁵、BioEn¹²⁶以及元推断⁹⁶。这些方法通过两种方式实现该原理：一种是对现有构象进行事后重新加权，另一种是在模拟过程中施加偏置力以引导采样过程。此类技术对于内在无序蛋白（IDPs）尤为有效——这类蛋白质的天然状态具有广泛的结构异质性，而在进行模型精修时必须保留这种异质性。

第二类方法称为最大简约法（MP），其旨在从庞大的构象集合中筛选出一个最小的子集，该子集能够充分解释实验数据，从而最小化结构复杂度。与ME方法不同，MP方法旨在通过选择或重新加权分子结构来简化构象集合——即在最小化构象集合规模的同时，最大化其与实验观测结果的一致性。这些方法（例如 EOM¹²⁷和 ASTEROIDS¹²⁸）在仅存在少数主导构象状态时尤为适用；因此，它们最适合用于构象异质性有限的低熵系统——这类系统无需采用统计系综进行描述。

总体而言，这些整合性策略使研究人员能够构建既符合物理现实、又经实验验证的结构集合模型，从而弥合了在无序蛋白质研究中模拟与实验之间的鸿沟。关于实验数据整合的代表性技术，详见**补充表3**。

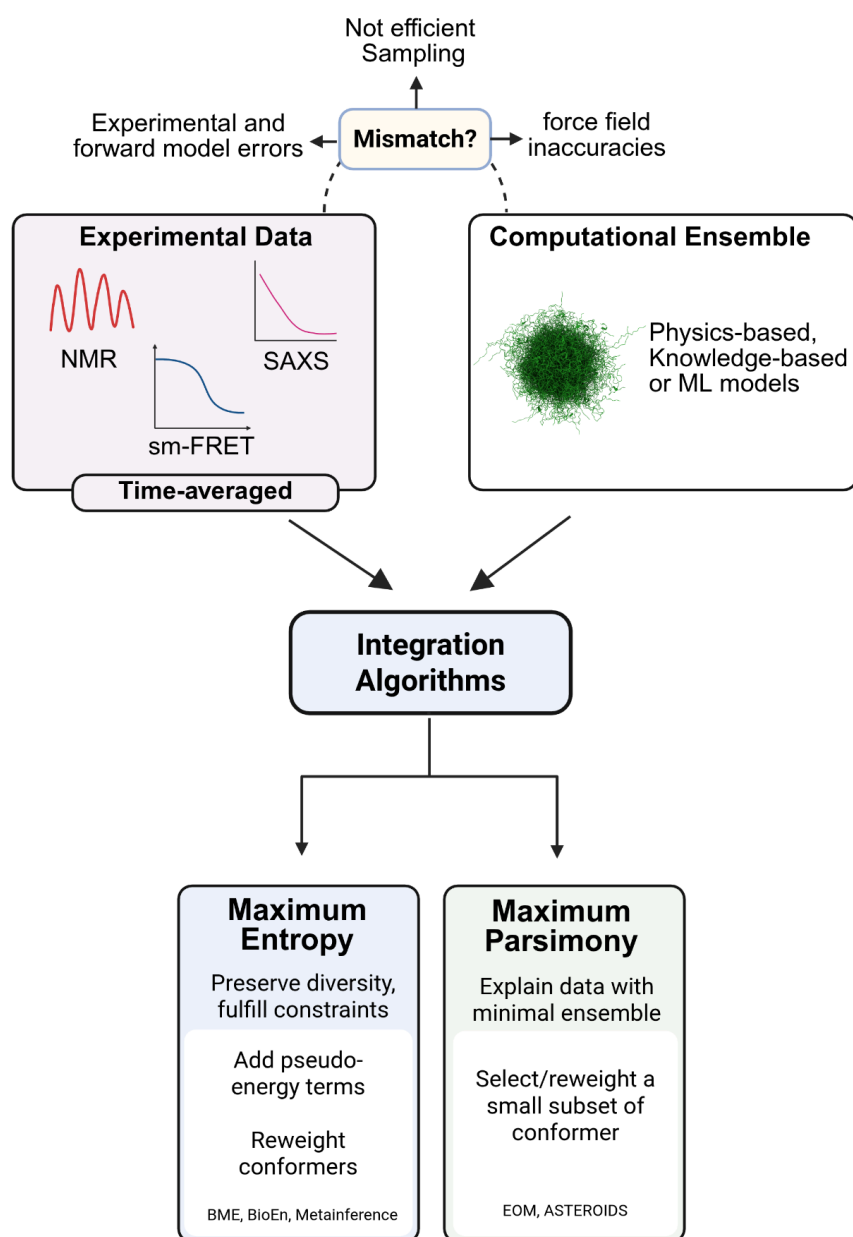


Fig. 3. Overview of integrative modeling strategies for intrinsically disordered proteins (IDPs). Experimental methods like NMR, SAXS, and smFRET provide averaged data across many conformations, while computational models generate detailed structural ensembles. When the two disagree—due to limited sampling, force field inaccuracies, or experimental noise—different strategies can be used to improve agreement. These include maximum entropy (ME) methods that adjust structural ensemble weights, maximum parsimony (MP) methods that generate conformational collections by selecting a minimal subset of structures. These strategies help align computational models with experimental data.

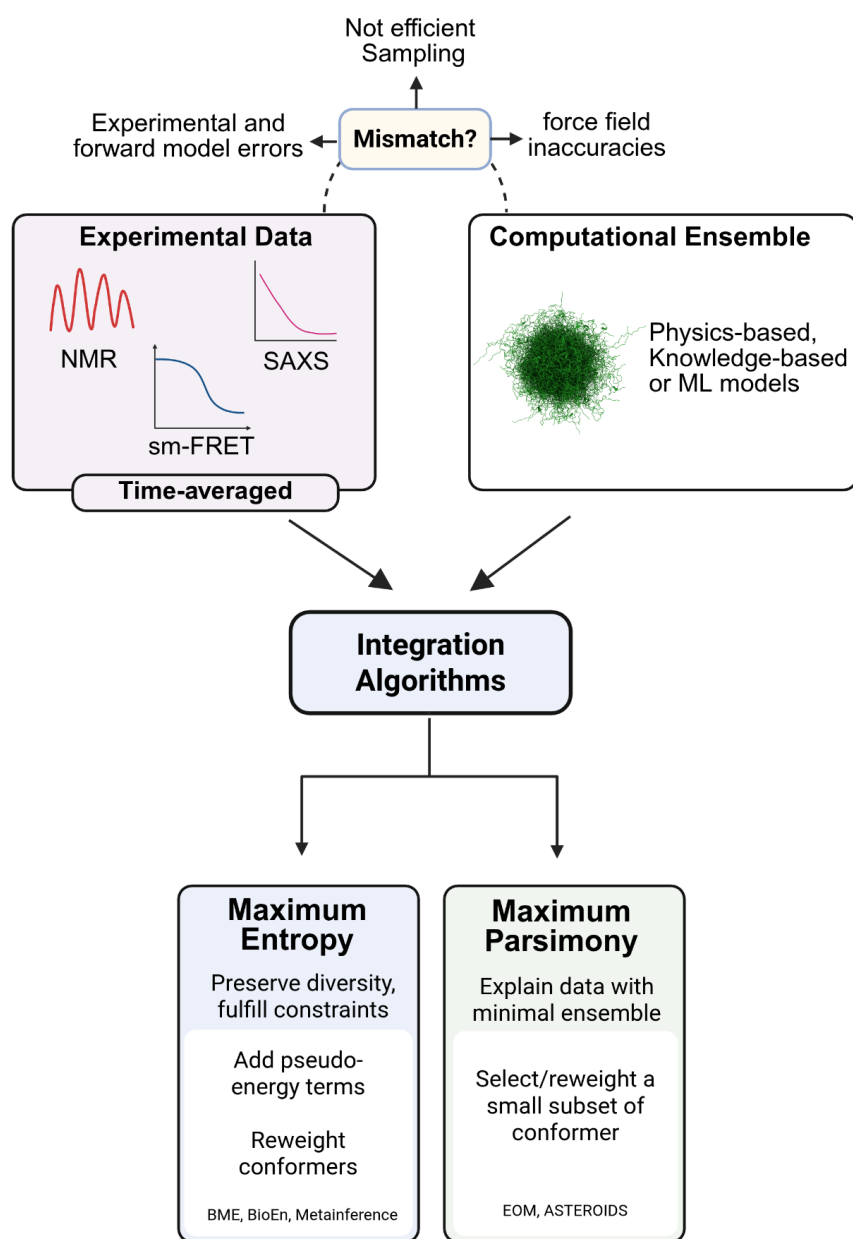


图 3。基于内在无序蛋白质（IDPs）的整合建模策略概述。实验方法（如 NMR、SAXS 和 smFRET）可提供涵盖多种构象的平均化数据，而计算模型则可生成详细的结构系综。当这两种方法所得结果不一致时——原因可能是采样不足、力场精度问题或实验噪声——可采用不同的策略来提高两者的一致性。这些方法包括：最大熵（ME）方法（用于调整结构系综权重）以及最大简约（MP）方法（通过选择最小结构子集来生成构象集合）。这些策略有助于使计算模型与实验数据相匹配。

Hybrid Computational techniques

Some computational methods for modeling IDPs leverage complementary modeling strategies. These hybrid approaches (**Supplementary Table 2**) integrate complementary ensemble generation techniques, or couple ensemble generation directly with experimental data integration strategies. One strategy involves combining physics-based and knowledge-based approaches to enhance sampling efficiency and structural diversity. For example, FastFloppyTail¹²⁹ employs fragment-based backbone sampling alongside energy-based refinement. Other techniques couple machine learning with physics-based modeling, such as AlphaFold–Metainference¹³⁰ and bAles¹⁰⁷ which refines deep learning predictions from AlphaFold via Bayesian ensemble inference and molecular simulation. A different class of hybrid methods links ensemble generation with experimentally data-driven refinement. For instance, DynamICE¹¹⁹ integrates machine learning with maximum entropy principles to iteratively optimize ensemble structures based on NMR data. These flexible and composable pipelines offer practical advantages when addressing the complexity of IDP ensembles, particularly in cases where no single modeling paradigm is sufficient.

Open challenges in conformational ensemble generation

Despite recent advances, several challenges remain in generating conformational ensembles¹²³. First, force field accuracy continues to be a critical limitation. While parametrizations using experimental data and quantum chemistry calculations are continuously improving their accuracy, current force fields still rely on rather simple functional forms established decades ago, and fail to describe subtle interactions, such as those arising from polarization effects, especially in complex or highly flexible biomolecules. Progress in developing more accurate force fields may require closer collaboration between experimental and computational researchers and the development of top-down approaches, including the adoption of machine-learning methods^{109,120,131}.

Second, the optimal integration of experimental data into simulations is still an open problem^{123,96}. Many forward models used to translate experimental data into simulation restraints are limited by their assumptions and approximations. Accounting for the ensemble-averaged nature of most experimental observables on IDPs, such as those derived from NMR or smFRET, or the effect of the hydration for SAS, is critical for accurate modeling. Robust frameworks that address uncertainties in both experimental data and forward models are essential to bridge this gap^{96,123}. In the absence of ground-truth conformational ensembles, such frameworks require experimental data from orthogonal techniques and for different proteins to self-consistently select, refine, and validate forward models^{97,99,132}. These experimental data are typically insufficient to describe a unique conformational ensemble, with the final result dependent on the computational model used to generate the ensemble.

Third, the efficient sampling of conformational space also remains a significant bottleneck. Insufficient sampling can lead to incomplete or biased conclusions, making it crucial to assess convergence rigorously. Advancements in some techniques, such as accelerated molecular

混合计算技术

部分用于模拟内在无序蛋白 (IDPs) 的计算方法采用了互补的建模策略。这些混合建模方法 (附表 2) 整合了互补的集合生成技术, 或直接将集合生成与实验数据整合策略相结合。其中一种策略是结合基于物理的与基于知识的建模方法, 以提高采样效率和结构多样性。例如, Fast FloppyTail¹²⁹ 采用了基于片段的主链采样方法结合基于能量的优化方法。其他技术则将机器学习与基于物理的建模相结合, 例如 AlphaFold–Metainference¹³⁰ 以及 bAles¹⁰⁷——后者通过贝叶斯集成推断和分子模拟对 AlphaFold 的深度学习预测结果进行优化。另一类混合方法则将集合生成与基于实验数据驱动的优化相结合。例如, DynamICE¹¹⁹ 将机器学习与最大熵原理相结合, 基于 NMR 数据对集成结构进行迭代优化。这些灵活且可组合的数据处理流程在应对 IDP 集合所蕴含的复杂性时具有显著的实际优势, 尤其是在单一建模范式不足以应对的情况之下。

构象集合生成领域的开放性挑战

尽管近期取得了诸多进展, 但在构建构象集合¹²³方面仍存在若干挑战。首先, 力场的准确性仍然是一个关键性的制约因素。尽管基于实验数据和量子化学计算的参数化方法其准确性正在持续提高, 但当前的力场仍依赖于数十年前确立的相对简单的函数形式, 因而难以描述诸如由极化效应引发的微妙相互作用——尤其是在复杂或高度柔性生物分子中。开发更精确力场所取得的进展, 可能需要实验研究者与计算研究者之间开展更紧密的合作, 以及开发自上而下的研究方法——包括采用机器学习方法^{109,120,131}。

其次, 如何将实验数据最优地整合到模拟中仍是一个尚未解决的问题^{123、96}。许多用于将实验数据转化为模拟约束条件的前向模型均受限于其自身的假设与近似处理。对于 IDPs 而言, 考虑大多数实验观测值 (例如源自 NMR 或 smFRET 的观测值) 所具有的系综平均特性, 或考虑 SAS 中水合作用的影响, 对于实现精确建模至关重要。能够同时处理实验数据与前向模型中不确定性的鲁棒框架对于弥合这一差距至关重要^{96,123}。在缺乏真实构象集合的情况下, 此类框架需利用来自不同 orthogonal 技术的实验数据, 对不同蛋白质的前向模型进行自洽的筛选、优化与验证^{97,99,132}。这些实验数据通常不足以描述一个独特的构象集合, 其最终结果取决于用于生成该构象集合的计算模型。

第三, 构象空间的高效采样仍是一个重要的瓶颈问题; 采样不足可能导致结论不完整或存在偏倚, 因此对收敛性进行严格评估至关重要。某些技术 (如加速分子动力学和马尔可夫状态模型

dynamics and Markov state models^{133,134}, offer potential solutions but require widespread adoption and careful validation.

Conformational ensemble validation and comparison

Comparison of conformational ensembles

The criteria and metrics for validating and comparing structures of well-folded proteins determined by X-ray crystallography, NMR spectroscopy, cryo-EM/ET, integrative approaches, or predicted by ML models, like AlphaFold¹³⁵, are well established. Validation checks include evaluating bond angles, torsion angles, and ensuring that the protein adheres to known stereochemical properties. Tools like MolProbity¹³⁶ are often used for these validations. For comparison, the most widespread approach is based on structural alignment and quantitative RMSD (Root Mean Square Deviation) analysis¹³⁷. Other scores to quantify the differences between protein structures (usually between predictions and experimental structures), such as the GDT (Global Distance Test)¹³⁸ or the TM-score (Template Modeling Score)¹³⁹, are frequently used, in particular to assess predictions in CASP (Critical Assessment of Structure Prediction)¹⁴⁰.

In contrast, the structural analysis of conformational ensembles of IDPs is more complex. As most experimental methods (Table 1) provide averaged values of measurements, these are hard to connect to usual structural descriptors, and in particular to atomic coordinates. This represents a great challenge for the evaluation of computational methods that generate conformational ensembles as a weighted set of conformers relying on atomistic or coarse-grained models^{95,141}, as those deposited in the Protein Ensemble Database (PED)⁹¹.

The comparison of conformational ensembles, however, serves multiple purposes. The determination of an IDP ensemble from sparse experimental data is a mathematically underdetermined problem, leading to multiple solutions. The comparison of these ensembles may be critical for validating them (see next section), and/or for deriving mechanistic insights by identifying functional correlates (e.g. binding regions, dynamic linkers), comparing homologous IDPs and assessing the influence of solution conditions. As conformational ensemble comparison methods for folded proteins¹⁴² are difficult to generalize to IDPs, various approaches have been developed in recent years. A conformational ensemble can be represented by a single scalar that quantifies its conformational heterogeneity¹⁴³, and differences in this value can be used to compare two conformational ensembles. While this metric is easy to interpret, it offers only a limited view of the differences between conformational ensembles and lacks the resolution to capture more detailed structural or dynamical variations. A more sophisticated approach involves analyzing the differences in distance distributions of individual residue pairs and assessing their statistical significance between two conformational ensembles¹⁴⁴. However, given the vast conformational variability of IDPs, it is essential to consider both average properties and their distributions. Indeed, reducing IDP conformational descriptors to their mean can lead to significant information loss. Several methods have been proposed for extending

^{133,134}) 的进展提供了潜在的解决方案, 但这些技术尚需得到广泛采用并经过严谨验证。

构象集合验证与比较

构象集合比较

用于验证和比较通过X射线晶体学、核磁共振波谱学、冷冻电镜/电子断层扫描、整合研究方法或基于机器学习模型(如AlphaFold¹³⁵)预测所得的正确折叠蛋白质结构的准则与评估指标已得到广泛认可。验证检查包括评估键角与扭转角, 以及确保蛋白质符合已知的立体化学特性; 此类验证通常使用MolProbity¹³⁶等工具进行。作为对比, 目前最广泛采用的方法是基于结构比对与定量RMSD(均方根偏差)分析¹³⁷。其他用于量化蛋白质结构间差异(通常为预测结构与实验结构之间的差异)的评分方法, 例如GDT(全局距离检验)¹³⁸或TM-score(模板建模评分)¹³⁹, 常被广泛采用, 尤其用于在CASP(结构预测关键评估)¹⁴⁰中对预测结果进行评估。

相比之下, 内在无序蛋白(IDPs)构象集合的结构分析则更为复杂。由于大多数实验方法(表1)所提供的均为测量值的平均值, 因此这些数据难以与常规结构描述符——尤其是原子坐标——相联系。这对于评估那些基于原子级或粗粒化模型^{95,141}生成构象集合(即由多个构象异构体组成的加权集合)的计算方法而言, 是一项重大挑战; 此类构象集合例如已收录于蛋白质集合数据库(PED)⁹¹中。

然而, 构象集合的比较具有多重用途。基于稀疏实验数据确定无序蛋白(IDP)的构象集合是一个数学上欠定问题, 因而存在多个可能的解; 对这些构象集合进行比较对于验证其可靠性(参见下一节)或通过识别功能相关性(例如结合区域、动态连接区)、比较同源无序蛋白以及评估溶液条件的影响来获取机制性见解都至关重要。由于针对折叠蛋白质的构象系综比较方法¹⁴²难以直接推广至无序蛋白质(IDPs), 近年来人们开发了多种相应方法。构象系综可由一个单一标量来表示, 该标量用于量化其构象异质性¹⁴³; 而这两个构象系综之间的差异可利用该标量值进行比较。尽管该指标易于理解, 但其仅能有限地反映构象集合之间的差异, 且缺乏捕捉更精细结构或动力学变化的分辨率。一种更为精细的方法是分析单个残基对之间距离分布的差异, 并评估这两个构象集合之间这些差异的统计学显著性¹⁴⁴。然而, 鉴于内在无序蛋白(IDPs)具有巨大的构象变异性, 同时考虑其平均性质及其分布至关重要。事实上, 将IDP的构象描述符简化为其均值可能会导致显著的信息损失。已有多种方法被提出, 用于将比较结构分析扩展到分布框架中, 即利用各种

comparative structural analyses to a distributional framework, using various similarity measures to compare probability distribution functions, such as the Kullback-Leibler and the Jensen-Shannon divergence^{142,144,145}, Hellinger distance¹⁴⁶ or the Wasserstein distance¹⁴⁷. Among these, the Wasserstein distance is particularly appealing for its incorporation of the geometry of the underlying space, though it is more computationally demanding than the others. In this framework, different descriptors can be used to characterize conformations inside each ensemble and to compute pairwise distances. The most natural one is to use atomic coordinates and RMSD¹⁴², however, the required structural superposition hinders its application to IDPs. To circumvent this problem, other descriptors such as torsion angles¹⁴⁸ or other local structural properties¹⁴⁵ can be considered. For a more comprehensive comparison, local and global structural descriptors can be taken into account in a unified way¹⁴⁷. Other approaches for the comparative analysis of conformational ensembles, such as methods incorporating topology concepts¹⁴⁹ or the identification of local energy traps at the residue level¹⁵⁰, have been explored. **Supplementary Table 4** summarizes conformational ensemble comparison methods.

Validation of conformational ensembles

As discussed in the previous sections, there are several ways — experimental, computational, or integrated — of determining conformational ensembles of IDPs. The comparison of alternative solutions is not trivial, and it also impinges upon our ability to select for the correct ones that reflect all important structural aspects of the actual conformational ensembles. Although it is very challenging to know the accuracy and precision of a conformational ensemble, objective criteria to assess its quality can be defined. In general, a conformational ensemble can be validated by experimental data using orthogonal experimental techniques, pointing toward the use of integrative approaches.

The general path to validation begins with ascertaining that the IDP ensemble adheres to simple physicochemical and stereochemical rules, and it complies with all ensemble-related experimental data. That is, the first step is to check if all conformations in the ensemble comply with measures of bond lengths and torsion angles, location on the Ramachandran map, and avoidance of atomic clashes. These are all conditions not always currently met in PED ensembles⁹¹.

The next step is to check the adherence of the ensemble to mean global structural parameters derived from experimental techniques, primarily SAS and smFRET, or calculated from sequence by applying polymer-physical principles^{4,6,46}, such as the R_g , end-to-end distance, geometric deviation from a perfect sphere (asphericity) and scaling exponents (ν)^{151–154}. As such global parameters may be compromised when secondary structures or long-range contacts are present, one should proceed from global descriptors to more local conformational parameters, such as short- and medium-range distance information (coming from smFRET and PRE), descriptors based on inter-residue contact probabilities^{155,156}, secondary-structure content (as reported by CD or NMR), and local residue-specific conformational preferences (Ramachandran torsion angles ϕ and ψ), which can be derived primarily from NMR data (CSs, RDCs, NOEs).

相似性度量对概率分布函数进行比较，例如 Kullback-Leibler 散度和 Jensen-Shannon 散度^{142,144,145}、Hellinger 距离¹⁴⁶或 Wasserstein 距离¹⁴⁷。其中，Wasserstein 距离因其能够融入底层空间的几何结构而尤为引人注目，尽管其计算复杂度高于其他距离度量方法。在该框架下，可采用不同的描述符来表征每个构象集合中的构象，并计算两两距离。最自然的方法是采用原子坐标和 RMSD¹⁴²，然而，该方法对结构叠加的要求限制了其在内在无序蛋白（IDPs）中的应用。为解决这一问题，可考虑采用其他描述符，例如扭转角¹⁴⁸或其他局部结构特性¹⁴⁵。为了进行更全面的比较，可将局部与全局结构描述符以统一的方式纳入考量¹⁴⁷。此外，还探索了其他用于构象集合比较分析的方法，例如结合拓扑学概念的方法¹⁴⁹或在残基水平上识别局部能量陷阱的方法¹⁵⁰。**附表 4**汇总了各类构象集合比较方法。

构象集合的验证

如前文各节所述，存在多种确定内在无序蛋白（IDPs）构象集合的方法——包括实验法、计算法或综合法。对不同解的比较并非易事，这同时也影响着我们筛选出能够反映实际构象集合中所有重要结构特征的正确解的能力。尽管确定构象集合的准确度与精密度极具挑战性，但可制定客观的评价标准来评估其质量。总体而言，构象集合可通过采用正交实验技术的实验数据进行验证，这提示应采用综合性研究方法。

验证过程的总体路径始于确认 IDP 系综符合简单的物理化学及立体化学规则，并满足所有与系综相关的实验数据要求。也就是说，第一步是检查系综中的所有构象是否符合键长与扭转角的测量值、在 Ramachandran 图上的位置以及是否避免了原子间碰撞。然而，这些条件在目前的 PED 系综中并不总是得到满足⁹¹。

下一步是检验该集合对源自实验技术（主要是 SAS 和 smFRET）或通过应用聚合物物理原理^{4,6,46}从序列计算得出的全局结构平均参数的符合程度；这些参数包括 R_g （端到端距离）、与理想球体的几何偏差（非球面度）以及缩放指数（ α ）^{151–154}。由于当存在二级结构或长程相互作用时，这些全局参数可能受到影响，因此应从全局描述符过渡到更局部的构象参数——例如短程与中程距离信息（源自 smFRET 和 PRE）以及基于残基间接触概率的描述符^{155,156}。二级结构含量（通过圆二色性（CD）或核磁共振（NMR）技术测定）以及局部残基特异性构象偏好性（Ramachandran 扭转角 φ 与 ψ ），这些参数主要可从核磁共振数据（CSs、RDCs、NOEs）中导出。

The next step is to check for the robustness of a conformational ensemble, by assessing how it depends on particular data points. A sign of limited accuracy is large alterations of the ensemble upon removing part of the input data^{123,157}. On the more positive side, the quality of the conformational ensemble can be ascertained if the ensemble can predict independent experimental observables¹⁵⁸, as demonstrated in two different settings. In one study, representative ensembles of α -synuclein (140 amino acids) and tau protein (441 amino acids) were derived from NMR and SAXS data. These conformational ensembles were cross-validated by selectively omitting portions of the data during the ensemble selection procedure and subsequently back-calculating the omitted data from the resulting ensembles¹⁵⁹. This approach was shown to lead to more accurate predictions than those from a statistical coil model. In another similar experiment, the structural ensemble of the disordered N-terminus of measles virus phosphoprotein (110 amino acids) was approached by a variety of NMR, SAXS and smFRET data, with particular emphasis on the interplay of NMR paramagnetic relaxation enhancement (PRE) and smFRET data expectedly providing similar constraints on medium- and long-range distance distributions¹²⁸. Interestingly, the conformational ensembles calculated by leaving one type of data out – PRE or smFRET – had some power in reproducing the other, but the full complement of data was required for reproducing all aspects of the ensemble.

A more abstract structural feature to judge for validation is if the conformational ensemble has some biological/functional relevance. If an IDP interacts with other molecules (e.g., proteins, DNA, RNA), one may assess its ability to recapitulate the experimentally observed binding interfaces and binding modes. A particular feature toward this goal is that IDPs often function by molecular recognition undergoing disorder to order transition upon binding which can assume an unbound structural state similar to their partner bound conformation¹⁶⁰. Capturing this feature by the conformational ensemble is a sign of its functional relevance^{144,161}. Recently, AlphaFold has been shown to reliably predict relevant binding modes of disordered regions¹⁶². There are also cases where the global dimensions of linker regions have been shown to be evolutionarily conserved^{4,5}. Additionally, correlations have been observed between IDP compaction and gene-ontology (GO) protein function descriptors⁴.

In a similar vein, one might check for disease relevance. If an IDP is involved in disease, one may consider if its conformational ensemble provides insights into disease mechanisms and/or potential therapeutic targeting. To this end, one may ask if *in silico* docked small molecules do experimentally bind, whereas extending these to broader structure-activity-relationships (SAR) of molecular analogues has even more confirmatory power^{163–165}.

As suggested above, determining conformational ensembles from sparse experimental data is a mathematically ill-posed problem, which has multiple solutions¹²³. However, from an experimental point of view, the problem becomes well-posed if different conformational ensembles are experimentally indistinguishable^{4,123}.

Looking forward, we note that the conformational ensembles that we aim to compare and validate are determined mostly under *in vitro* conditions, and thus they may not adequately reflect the true structural preferences of the proteins in their physiological, cellular milieu. That is, the ultimate verification of a conformational ensemble is to validate it against cellular

下一步是通过评估构象集合对特定数据点的依赖程度，来检验该构象集合的稳健性。若在移除部分输入数据后，构象集合发生显著变化，则可视为其精度有限的标志^{123,157}；从积极角度来看，若该构象集合能够预测独立的实验可观测量¹⁵⁸，则可确认其质量——这一点已在两种不同的实验设置中得到验证。在一项研究中，研究人员基于核磁共振（NMR）和小角X射线散射（SAXS）数据构建了 α -突触核蛋白（140个氨基酸）与tau蛋白（441个氨基酸）的代表性构象集合。这些构象集合通过在集合筛选过程中选择性地剔除部分数据，随后根据所得集合对被剔除的数据进行反向计算的方式进行交叉验证¹⁵⁹。研究表明，与基于统计性线团模型所得的预测结果相比，该方法能够得出更准确的预测结果。在另一项类似的实验中，研究人员综合利用多种核磁共振（NMR）、小角X射线散射（SAXS）及单分子FRET（smFRET）实验数据来解析麻疹病毒磷蛋白（含110个氨基酸）无序N端的结构集合体；其中特别侧重于分析NMR顺磁弛豫增强（PRE）与smFRET数据之间的相互作用——正如预期，这两类数据对中程及长程距离分布均提供了相似的约束条件¹²⁸。有趣的是，通过剔除其中一种数据类型（PRE 或 smFRET）所计算出的构象集合，在重现另一种数据类型时具有一定效力；然而，要完整重现该构象集合的所有特征，则需要使用全部数据集。

另一个可用于验证的抽象结构特征是：该构象集合是否具有某种生物学或功能相关性。若该无序蛋白（IDP）与其它分子（例如蛋白质、DNA 或 RNA）发生相互作用，则可评估其再现实验观测到的结合界面与结合模式的能力。实现这一目标的一个重要特征在于：IDPs 通常通过分子识别机制发挥作用——即在结合过程中经历从无序态到有序态的转变，此时其可呈现一种与配体结合构象相似的未结合结构状态¹⁶⁰；通过构象系综捕捉这一特征，表明了该特征在功能上的重要性^{144,161}。近期研究表明，AlphaFold 能够可靠地预测无序区域相关的结合模式¹⁶²。此外，有研究发现连接子区域的全局维度在进化上是保守的^{4,5}。另外，研究还观察到非结构化蛋白（IDP）的压缩程度与基因本体论（GO）蛋白质功能描述符之间存在相关性⁴。

同样地，可探究该分子与疾病的关联性；若某种IDP参与了某种疾病的发病过程，便可考虑其构象集合是否能为揭示该疾病的发病机制和/或潜在的治疗靶点提供线索。为此，可验证通过*计算机模拟*对接得到的小分子是否在实验中确实能够结合；而将这一方法拓展至分子类似物更广泛的构效关系（SAR）研究，则具有更强的验证效力^{163–165}。

如上所述，基于稀疏实验数据确定构象集合是一个数学上不稳定的问题，该问题存在多个解¹²³。然而，从实验角度来看，若不同的构象集合在实验上无法区分，则该问题即变为稳定问题^{4,123}。

展望未来，我们注意到，本文旨在比较和验证的构象集合大多是在体外条件下确定的，因此它们可能无法充分反映蛋白质在生理或细胞微环境中的真实结构偏好。也就是说，对构象集合的最终验证应通过将其与细胞内结构数据进行比对来实现；例如，可通过细胞内核磁共振（NMR）或单

structural data, for example obtained by in-cell NMR or smFRET^{51,166}, which are still in their infancy. It has already been shown, however, that the structural disorder of several IDPs persists in mammalian cells and show behavior close to that recorded *in vitro*, e.g. α -synuclein investigated by NMR⁴¹, prothymosin α observed with smFRET⁵¹, or comparisons between *in vitro* and *in vivo* FRET measurements¹⁶⁷. Such approaches should be routinely invoked as a conclusive element of IDP ensemble validation. Another approach involves using analyses of sequence conservation signals to gauge the conformational properties of IDPs that are important for function in living organisms^{168–170}.

Roadmap for conformational ensemble generation

The determination of conformational ensembles of IDPs stands at a critical juncture, where methodological diversity is expanding rapidly but standardized practices for validation, comparison, and application remain elusive. To address these challenges, we propose a four-pronged roadmap aimed at shaping a rigorous and accessible future for IDP ensemble modeling. This includes: (1) establishing a benchmarking infrastructure to drive community-wide progress, (2) developing robust methods for quantifying and propagating uncertainty in ensemble generation and evaluation, (3) incorporating strategies for capturing non-equilibrium dynamics and kinetic features of disordered systems, and (4) offering practical guidelines for method selection.

1. IDP Ensemble Benchmarking Challenge (IDP-Bench)

A cornerstone for advancing the field is the creation of an open, transparent, and standardized benchmarking initiative. Inspired by the success of CASP¹⁷¹ in protein structure prediction and of CAID in the critical assessment of intrinsic disorder prediction¹⁷², we introduce the IDP Ensemble Benchmarking Challenge (IDP-Bench), a recurring, community-driven assessment platform specifically tailored to IDP structural ensembles. IDP-Bench will be structured as a four-phase biennial cycle aligned with CAID initiative (**Supplementary Fig. 1**: (i) selection and curation of high-quality, diverse experimental datasets by an independent committee, (ii) submission of ensemble predictions by participating groups, (iii) automated, multi-metric scoring against withheld validation data, and (iv) continuous public dissemination of results via a dedicated website. The first edition will evaluate predictions for well-characterized IDPs, such as α -synuclein, Sic1, the K18 repeat domain of tau and full-length tau, p27Kip1, and the measles-virus NTAIL domain, each with extensive public datasets (SAXS, PRE, RDC, smFRET, HDX-MS). These proteins collectively span different sequence features, biological functions, and levels of experimental accessibility, and none has yet been evaluated under a uniform blind protocol.

Crucially, scoring metrics will assess both global (e.g., χ^2 for SAXS profiles, KL divergence for $P(r)$) and local agreement (e.g., PRE RMSE, RDC Q-factor, smFRET Wasserstein distance) (see **Supplementary Table 5**). These will be combined into a weighted composite score, with full scoring scripts and error models released as open-source tools.

A blind component will be introduced further in IDP-Bench, e.g., SAXS or NMR data acquired under a given set of conditions, in order to test true predictive capability.

分子荧光共振能量转移 (smFRET) 技术^{51,166}获取此类数据——这些技术目前尚处于发展初期。然而, 已有研究证实, 多种内在无序蛋白 (IDPs) 的结构无序性在哺乳动物细胞中依然存在, 且其行为与体外实验中记录到的行为十分接近——例如通过核磁共振 (NMR) 对 α -突触核蛋白进行的研究⁴¹、利用单分子FRET技术对前胸腺素 α 进行的观测⁵¹, 或体外与体内FRET测量结果之间的对比研究¹⁶⁷。此类方法应作为IDP整体构象验证中的决定性依据而常规采用。另一种方法是利用序列保守性信号分析来评估IDPs的构象特性, 这些特性对生物体内的功能至关重要^{168–170}。

构象系综生成路线图

IDP构象集合的确定研究正处于一个关键节点: 尽管相关研究方法的多样性正在迅速扩大, 但针对验证、比较及应用的标准化操作规范仍不完善。为应对这些挑战, 我们提出一项包含四个方面的路线图, 旨在为IDP集合建模领域构建一个严谨且易于使用的未来。这包括: (1) 建立基准对比基础设施以推动整个社区的协同发展; (2) 开发 robust 的方法用于量化及传递集成模型生成与评估过程中的不确定性; (3) 整合用于捕捉无序系统非平衡动力学及动力学特征的策略; (4) 提供针对方法选择的实用指导方针。

1. IDP集成基准测试挑战赛 (IDP-Bench)

推动该领域发展的基石之一, 是创建一个开放、透明且标准化的基准测试计划。受 CASP¹⁷¹在蛋白质结构预测领域取得的成功, 以及 CAID 在内在无序性预测关键性评估中所取得的成就¹⁷²的启发, 我们推出了 IDP 集合基准测试挑战赛 (IDP-Bench) ——这是一个由社区主导、定期开展的评估平台, 专为 IDP 结构集合而设计。IDP-Bench 将采用一个与 CAID 计划相契合的四阶段双年度循环模式 (附图 1: (i) 由独立委员会筛选并整理高质量、多样化的实验数据集; (ii) 参与研究团队提交集成预测结果; (iii) 对独立保留的验证数据集进行自动化、多指标评分; (iv) 通过专用网站持续公开研究成果)。第一版将对特征明确的内在无序蛋白 (IDPs) ——例如 α -突触核蛋白、Sic1、tau蛋白的K18重复结构域及全长tau蛋白、p27Kip1以及麻疹病毒NTAIL结构域——的预测结果进行评估; 上述蛋白均拥有规模庞大的公开实验数据集 (包括SAXS、PRE、RDC、smFRET、HDX-MS等)。这些蛋白质共同涵盖了不同的序列特征、生物学功能以及实验可及性水平, 且迄今尚未有任何一种蛋白质在统一的盲法实验方案下进行过评估。

至关重要的是, 评分指标将同时评估全局一致性 (例如: SAXS 比对曲线采用 χ^2 , P(r) 采用 KL 散度) 与局部一致性 (例如: PRE RMSE、RDC Q-因子、smFRET Wasserstein 距离) (参见补充表 5)。这些指标将综合计算为一个加权综合评分, 完整的评分脚本及误差模型已作为开源工具发布。

在 IDP-Bench 中将引入盲数据集 (例如在特定实验条件下获取的 SAXS 或 NMR 数据), 以检验其真实的预测能力。

Altogether, this initiative lays the foundation for a sustainable benchmarking ecosystem—one that promotes transparency, drives methodological innovation, and enables long-term tracking of progress in IDP ensemble prediction.

2. Uncertainty in conformational ensemble determination

Benchmarking conformational ensembles of IDPs requires not only standardized data and scoring metrics, but also a rigorous framework for handling uncertainty. We therefore plan to integrate uncertainty quantification directly into the IDP-Bench framework, making it a foundational criterion for evaluating the quality of the conformational ensembles.

Supplementary Fig. 2 summarizes the error landscape in the determination of IDP ensembles, focusing on key experimental observables. It serves as a baseline guide for experimental reproducibility and model fitting. We also should consider forward-model error in our variance estimates. For instance, chemical shift back-calculations using SPARTA+¹⁷ yield a standard deviation of about 1 ppm for backbone C α predictions on its training dataset¹⁷³, and SAXS profile predictions evaluated against benchmark consensus data reveal that even widely used tools like CRY SOL¹⁷⁴ can show systematic deviations due to differences in hydration layer modeling and background correction¹⁷⁵. Where empirical estimates are lacking, we advocate bootstrapping over modeling parameters (e.g., fluorophore grids, rotamer libraries) to obtain robust confidence intervals. The IDP-Bench scoring pipeline will explicitly output both standard and scoring metrics: reduced χ^2 , residual distributions, and Bayes factors. These help distinguish between genuine fits and artifacts of overfitting. To enforce transparency, future participants will submit a standardized reporting checklist detailing: (i) experimental data with point-wise uncertainties, (ii) forward-model error sources and propagation strategies, (iii) bootstrap/posterior distributions for fitted parameters, and (iv) treatment of outliers.

By embedding uncertainty quantification into every stage, from experimental input to scoring and reporting, we ensure that ensemble accuracy is assessed not only in absolute terms, but relative to the data's inherent noise and modeling limitations.

3. Dynamic properties and non-equilibrium behavior of IDPs

Non-equilibrium dynamical processes represent a promising but still nascent area in IDP structural biology. Many IDPs function through transient interactions and conformational switching, which are not adequately captured by equilibrium models. A major challenge is the acquisition of time-resolved experimental data, but recent advances in techniques such as time-resolved X-ray crystallography with free-electron lasers¹⁷⁶, real-time NMR¹⁷⁷, time-resolved and single-molecule FRET¹⁷⁸, time-resolved HDX-MS¹⁷⁹ and SAXS¹⁸⁰, and liquid-state electron microscopy¹⁸¹ are beginning to enable such studies. These methods are increasingly being used to provide structural restraints for simulations that aim to reconstruct the time evolution of IDP ensembles.

To model these processes, recent approaches apply frameworks based on maximum entropy and its dynamical extension, maximum caliber. These allow for the inference of IDP dynamics consistent with both experimental data and physical principles, enabling the modeling of kinetic pathways and transient conformational states^{182,183}. While still in early development, this

总体而言，这一倡议为构建一个可持续的基准测试生态系统奠定了基础——该生态系统旨在促进透明度、推动方法论创新，并实现对IDP集成预测进展的长期跟踪。

2. 构象系综测定中的不确定性

对内在无序蛋白（IDPs）的构象集合进行基准测试，不仅需要标准化的数据和评分指标，还需要一个用于处理不确定性的严谨框架。因此，我们计划将不确定性量化直接整合到 IDP-Bench 框架中，使其成为评估构象集合质量的基础性准则。**附图 2**概述了在确定 IDP 构象集合过程中所涉及的误差分布情况，重点关注关键的实验观测量。该指南可作为实验可重复性与模型拟合的基准参考。在进行方差估计时，我们还应考虑前向模型所产生的误差。例如，采用 SPARTA+¹⁷进行的化学位移反演计算显示，在其训练集¹⁷³上进行主链 $C\alpha$ 坐标预测时，标准差约为 1 ppm；而基于基准共识数据对 SAXS 比峰预测结果进行评估后发现，即便是像 CRY SOL¹⁷⁴这样广泛使用的工具，也可能因水合层建模与背景校正方法的差异而产生系统性偏差¹⁷⁵。在缺乏经验性估计值的情况下，我们建议采用自助法对模型参数进行估计（例如）。（荧光团网格、旋转异构体库等）用于获取稳健的置信区间。IDP-Bench评分流程将明确输出标准指标与评分指标：简化 χ^2 、残差分布以及贝叶斯因子。这些指标有助于区分真实拟合与过拟合所致的人为假象。为确保透明度，未来参与方需提交一份标准化的报告清单，详细说明：(i) 带有点估计不确定度的实验数据；(ii) 前向模型的误差来源及误差传播策略；(iii) 拟合参数的自助法/后验分布；以及 (iv) 异常值的处理方法。通过将不确定性量化方法嵌入到从实验输入到评分与报告的每个环节中，我们确保对集成模型的准确度进行评估时，不仅采用绝对值衡量，还将其与数据固有的噪声及建模局限性进行相对比较。

3. IDPs的动态特性与非平衡行为

非平衡动力学过程是IDP结构生物学中一个前景广阔但尚处于起步阶段的研究领域。许多IDP通过瞬时相互作用与构象转换来发挥功能，而这些过程无法被平衡态模型充分描述。一个主要挑战在于获取时间分辨实验数据，但近年来诸如利用自由电子激光器进行的时间分辨X射线晶体学¹⁷⁶、实时核磁共振（NMR）¹⁷⁷、时间分辨及单分子FRET技术¹⁷⁸、时间分辨HDX-MS¹⁷⁹与SAXS¹⁸⁰，以及液态电子显微镜技术¹⁸¹等技术的进展，正逐步为这类研究提供支持。这些方法正被越来越多地用于为旨在重建 IDP 系综时间演化的模拟提供结构约束。为对这些过程进行建模，近期的研究方法采用了基于最大熵及其动力学扩展——最大容量（maximum caliber）的理论框架。这些框架支持推断出既符合实验数据又符合物理原理的IDP动力学行为，从而能够对动力学反应路径及瞬时构象状态进行建模^{182,183}。尽管该研究方向尚处于早期

direction offers a path toward dynamic ensemble models that more closely reflect the time-dependent behavior of IDPs in their functional contexts.

4. Guidelines for method selection

Beyond standardization and validation, progress in IDP ensemble determination requires clear guidance for choosing the appropriate tools and strategies based on available data, biological questions, and computational resources. To this end, we propose a pragmatic decision framework for method selection, derived from the typical reasoning process of experienced practitioners. The guidelines begin by evaluating available experimental data. When no structural data are available, molecular dynamics simulations (e.g., coarse-grained models or ML-generated backbones) can offer baseline hypotheses. For systems constrained only by local observables (e.g., chemical shifts), we recommend knowledge-based or ML-based backbone generators coupled with reweighting. When both local and global data (e.g., smFRET, SAXS) are available, hybrid restrained simulations or Bayesian/maximum-entropy methods offer optimal balance between fidelity and generalizability. Next, the biological goal should guide model resolution. All-atom models are ideal for site-specific predictions (e.g., binding interfaces, mutational effects), while coarse-grained models excel in large-scale problems (e.g., phase behavior, scaling laws) where speed and statistical representativeness are key.

Finally, an important consideration in method selection is the available computational resources. For projects with limited resources, such as a desktop workstation, methods based on accelerated Monte Carlo sampling or knowledge-based generation may be most appropriate. Access to a modest GPU cluster enables more advanced techniques, including replica-exchange molecular dynamics or hybrid reweighting approaches. For users with access to high-performance computing infrastructure, all-atom replica simulations and machine learning-based ensemble refinement methods become feasible options.

These decision rules are captured in a simplified flowchart (**Supplementary Fig. 3**) and two supplementary tables: (i) ensemble generation tools cross-referenced with their input types, compute cost, and best-use cases (**Supplementary Table 2**), and (ii) data integration tools indicating whether they propagate uncertainty and accommodate forward-model variability (**Supplementary Table 3**).

Conclusion

In the face of growing interest in disordered proteins and the understating of their structural properties, this article establishes a comprehensive and practical framework to guide IDP ensemble determination, benchmarking, and interpretation. By integrating community expertise across experimental and computational domains, we chart a roadmap for IDP ensemble determination, emphasize the need for uncertainty quantification, and propose actionable benchmarking strategies. To implement this roadmap, we established the IDP Ensemble Benchmarking Challenge (IDP-Bench), following the examples of CASP and CAID, which greatly contributed to major advances in computational structural biology. This collaborative effort not only provides clarity on the current landscape of conformational ensemble modeling but also defines a shared vision for its future. As the field moves toward consensus and

发展阶段，但它为构建动态集成模型提供了途径，使这些模型能更准确地反映内在无序蛋白（IDPs）在其功能背景下的时变行为。

4. 方法选择指南

除标准化与验证工作外，IDP集成测定方法的进展还亟需明确的指导方针，以便根据现有数据、生物学研究问题及计算资源选择适宜的工具与策略。为此，我们基于经验丰富的从业者典型的推理过程，提出一个实用的方法选择决策框架。该指南首先对现有实验数据进行评估。当缺乏结构数据时，分子动力学模拟（例如粗粒化模型或基于机器学习生成的主链）可为研究提供基准假设。对于仅受局部可观测量（如化学位移）约束的体系，我们推荐采用基于知识或基于机器学习的主链生成器，并结合重加权方法。当同时具备局部与全局数据（例如 smFRET、SAXS）时，混合约束模拟或贝叶斯/最大熵方法可在保真度与泛化能力之间实现最优平衡。随后，应以生物学研究目标为导向来确定模型的精细程度：全原子模型适用于进行位点特异性预测（例如结合界面、突变效应），而粗粒度模型则在处理大规模问题时表现优异（例如）。（相态行为、标度律）——在这些领域中，速度与统计代表性至关重要。

最后，在方法选择时，一个重要的考量因素是可用的计算资源。对于资源有限的项目（例如使用台式工作站的项目），基于加速蒙特卡罗采样或基于知识的生成方法可能最为适宜；而若可使用中等规模的 GPU 集群，则能够采用更先进的技术，例如副本交换分子动力学或混合重加权方法。对于能够使用高性能计算基础设施的用户而言，全原子副本模拟及基于机器学习的集合优化方法便成为可行的选择。

这些决策规则汇总于一张简化的流程图（附图 3）以及两份补充表格中：(i) 集成生成工具及其输入类型、计算成本和最佳应用场景的交叉引用表（附表 2）；(ii) 数据集成工具是否传递不确定性及是否兼容前向模型变异性说明表（附表 3）。

结论

随着人们对无序蛋白质研究兴趣的日益增长以及对其结构特性认识的不断深入，本文构建了一个全面且实用的框架，旨在为无序蛋白质（IDP）集合体的确定、基准测试及结果解读提供指导。通过整合实验与计算领域内的跨学科专业知识，我们为IDP集合体的确定工作绘制了路线图，强调了进行不确定性量化的重要性，并提出了可落地的基准测试策略。为落实这一路线图，我们参照 CASP 和 CAID 的成功经验，设立了 IDP Ensemble Benchmarking Challenge（IDP-Bench）——这一举措为计算结构生物学领域的重大突破作出了重要贡献。该合作项目不仅清晰地揭示了当前构象系综建模领域的现状，更为该领域的发展描绘了共同的愿景。随着该领域逐渐形成共识并实现系统

interoperability, we hope this work will serve as both a reference and a call to action, catalyzing rigorous, reproducible, and biologically relevant studies of disordered protein conformational space.

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Author contributions

All authors contributed to the discussion and the initial draft of this manuscript. H.G wrote the final document with the help of the co-authors. All authors edited and refined the final manuscript. A.M.M and S.C.E.T coordinated the project.

Competing interests

K.L.-L. holds stock options in and is a consultant for Peptone Ltd. All other authors declare no competing interests.

Additional information

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间互操作性，我们希望这项工作既能作为参考指南，也能激发科研人员采取行动，从而推动对无序蛋白质构象空间开展严谨、可重复且具有生物学意义的研究。

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附加信息

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